

BASS as a Surrogate Model for Bayesian Optimization

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Introduction

This is where the introduction will go.

There is not a single theorem, lemma, or proposition in the whole paper. Have my senses taken leave of me? What, no asymptotics or results concerning the rate at which MARS approaches the “True Model” as the sample size goes to infinity? For one of the few times in its history, the Annals of Statistics has published an article based only on the fact that this may be a useful methodology. All is ad hoc; there is no maximum likelihood, no minimax, no rates of convergence, no distributional or function theory. Is nothing sacred? What kind of statistical science is this? My thanks go to the editor and the others involved in this sacrilegious departure.

- Leo Breiman, discussing the MARS model.

Chapter 1

Stochastic Processes

Before starting, we would like to note that sections of this chapter were constructed using different references as primary guides. The section on Markov chains primarily uses the references [Žitković, 2010] and [Hohendorff and Rosenthal, 2005].

Stochastic processes play a very important role in basically every part of this work. First, the basis of traditional Bayesian Optimization is Gaussian processes, which are a specific class of stochastic process in which all the points or vectors have a conjoined multi-normal distribution. This is not the only place where stochastic processes play a role, as BASS is a Bayesian method and modern Bayesian methods generally rely on simulation strategies such as Markov Chain Monte Carlo (MCMC) to numerically approximate complex posterior distributions. We use a particular class of MCMC algorithms called reverse jump MCMC when dealing with our alternate surrogate function, but that is detailed much later when we talk about the Bayesian Adaptive Regression Spline method.

This chapter develops the theoretical foundations required to construct and reason about Gaussian processes, with the goal of arriving at Gaussian process regression as a principled framework for supervised learning under uncertainty. The exposition follows a deliberate progression: we begin with the theory of Markov chains, establish the ergodic properties that justify their use in sampling-based inference, and then build toward Gaussian processes as a continuous generalization of the finite-dimensional Gaussian distribution. Central to this development is the role of the kernel function, which encodes prior assumptions about the structure of the process and determines the qualitative behavior of its realizations. The chapter concludes with Gaussian process regression, which is the mechanism by which uncertainty is quantified and function approximations are constructed in the traditional approach to Bayesian optimization.

1.1 General Definitions

All in all, stochastic processes are generally important to the content of this work so we include some important results and definitions, starting with the definition of a stochastic process itself, taken from [Žitković, 2010].

Definition 1. Let $\Theta \subseteq \mathbb{R}^+$. A stochastic process $\{X_\theta, \theta \in \Theta\}$ is a collection of random variables, indexed by a parameter θ , such that θ belongs to some index set Θ . When $\Theta = \mathbb{N}$ (or $\Theta = \mathbb{N}_0$), then it is called a discrete-time process, and if $\Theta = \mathbb{R}^+$, then it is called a continuous-time process.

To distinguish between a discrete and continuous time process, we use the index n for discrete-time processes and t for continuous-time processes. In all applications of this work the indexes refer to time, as the function spaces $\mathcal{X}$ on which the functions are evaluated are time dependent. This definition is very broad, as for example, if we take the case where $\Theta = \{1\}$, then the stochastic process $\{X_\theta, \theta \in \Theta\}$ is really just X_1 , a random variable. As with many other areas of mathematics though, the introduction of infinities in the values the indexes can take introduces a large amount of complexity, and things like vector distributions do not extend nicely to these new cases.

Another more formal definition of a stochastic process defines it as a function of the index and an element of the sample space, as follows:

Definition 2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space¹ where Ω is a sample space, $\mathcal{F}$ is a σ -algebra on Ω and $\mathbb{P} : \mathcal{F} \mapsto [0, 1]$ a probability measure. Taking Θ an arbitrary index space, a stochastic process is a function

$$X : \Omega \times \Theta \mapsto \mathbb{R}$$

such that for all $\theta \in \Theta$,

$$X_\theta : \omega \mapsto X(\omega, \theta) \equiv X_\theta : \Omega \mapsto \mathbb{R}$$

is a measurable function, which is equivalent to saying that X_θ is a random variable.

When ω is fixed, the map $\theta \mapsto X(\omega, \theta) : \Theta \rightarrow \mathbb{R}$ is called a **sample function**, or a **realization** of the stochastic process. Geometrically, each realization traces a trajectory which is a single path through the range of X . These types of trajectories are of the kind shown in Figure 1.1, which displays 100 simulated realizations of a Gaussian random walk, a more concrete example of these types of functions. A natural application of this framework

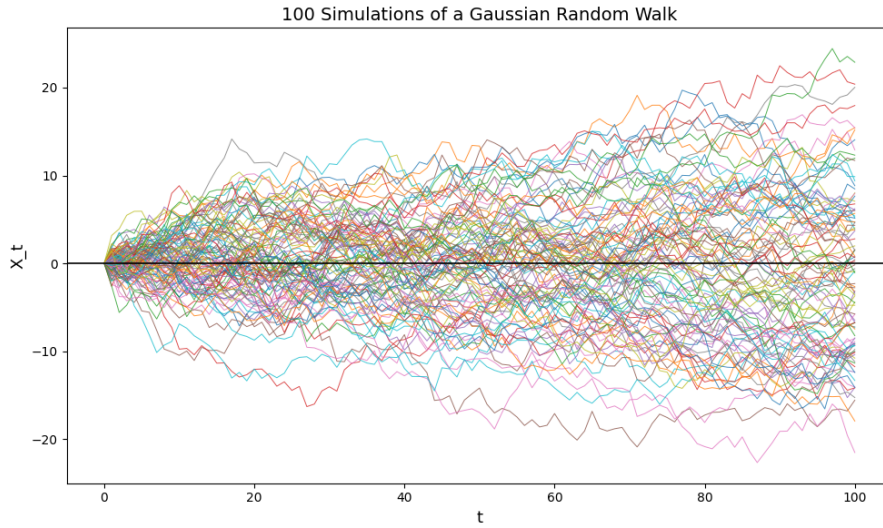


Figure 1.1: Random walks across a parameter t , where Θ is taken to be a time index set (with elements conventionally denoted t), and each realization represents one possible evolution of the process over time.

is **time series analysis**, where Θ is taken to be a time index set and each realization represents one possible evolution of the process over time.

As was mentioned earlier, the two main types of stochastic process that can be defined are ones where the realizations of the process occur in a discrete **time**, and the other is when they occur in a continuous fashion. It so happens that the two main structures relating to stochastic processes that we need to go into detail on are Markov chains, a discrete process, and Gaussian processes, which are continuous. We go in depth in the following sections into each of these types of stochastic processes, starting with Markov chains since Gaussian processes lead nicely onto the next chapter.

1.2 Markov Chains

Markov chains are ubiquitous in their applications, since they represent a relatively simple but very powerful concept: the mathematical structure on which we can define phase transitions of position changes that are governed by probabilistic rules. The fundamental property which will be explored later is that independently of the past, the only important factor in determining

¹Billingsley's classic book [Billingsley, 2012] has all the theoretical foundation of probability spaces and provides a thorough run down of these concepts.

the future state of the process is the current state of it. We begin with a formal definition and **build up the general theory of Markov Chains from there**.

Definition 3. Let $\{X_n, n \in \mathbb{N}_0\}$ be a stochastic process defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that

$$X_n : \omega \mapsto X(\omega, n) \equiv X_n : \Omega \mapsto \mathcal{S},$$

where $\mathcal{S}$ is a finite state space of cardinality k , such that $\mathcal{S} = \{s_1, s_2, \dots, s_k\}$. We call $\{X_n, n \in \mathbb{N}_0\}$ a Markov chain if

$$\mathbb{P}(X_n = s_n | X_{n-1} = s_{n-1}, X_{n-2} = s_{n-2}, \dots, X_0 = s_0) = \mathbb{P}(X_n = s_n | X_{n-1} = s_{n-1})$$

for all $n \in \mathbb{N}_0$ and all $s_0, s_1, \dots, s_k \in \mathcal{S}$ whenever the two conditional probabilities are well defined, this is, when $\mathbb{P}(X_{n-1} = s_{n-1}, \dots, X_0 = s_0) > 0$.

Intuitively, this means that the next state in the chain (**the sequence $\{X_n, n \in \mathbb{N}_0\}$**) is only dependent its current state, and that the history of how the chain got to the current state is irrelevant to the next step. This behaviour is known as the Markov property, and one very valuable insight that emerges from this behaviour is being able to completely determine all properties of the chain given only the initial state of it and the knowledge of how the chain behaves at all of its possible states.

The initial state or initial probability is self-explanatory, it tells us the expected behaviour of the first element in the chain, X_0 . Formally, we call $\mu^{(0)} = \{\mu_s^{(0)} : s \in \mathcal{S}\}$ with $\mu_s^{(0)} = \mathbb{P}(X_0 = s)$ the initial probability distribution of the process.

The other important component to fully describe a Markov chain is the transition probabilities from one state s_i to another s_j for all $s_i, s_j \in \mathcal{S}$. We identify this probability as

$$p_{i,j} = \mathbb{P}(X_n = s_j | X_{n-1} = s_i).$$

If we take all combinations of the states the chain can be in and map them to all other states, we get what is known as the transition matrix P , where each entry in the matrix denotes the probability of moving from one specific state to another. This matrix, because it is fundamentally a matrix made up of probabilities, has two interesting properties for the purposes of this work:

- $p_{i,j} \geq 0$ for all $i, j \in \{1, 2, \dots, n\}$,
- $\sum_{i=1}^n p_{i,j} = 1$ for all $j \in \{1, 2, \dots, n\}$.

We note that these two properties are the definition of a stochastic matrix, and therefore, by construction, Markov chain transition matrices are stochastic matrices. We have only described that we can fully determine the Markov chain from its initial state and its transition probabilities, but we have not shown this concept concretely. The following theorem ties these loose ends together and formally describes this property.

Theorem 1. *Let $\{X_n, n \in \mathbb{N}_0\} : \Omega \mapsto \mathcal{S}$ be a stochastic process defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathcal{S}$ a finite state space. Then, $\{X_n, n \in \mathbb{N}_0\}$ is a Markov chain if and only if there exists a stochastic matrix P such that*

$$\mathbb{P}(X_n = s_n | X_{n-1} = s_{n-1}, \dots, X_0 = s_0) = p_{n-1,n}$$

for all $n \in \mathbb{N}_0$ and all $s_0, s_1, \dots, s_n \in \mathcal{S}$ such that $\mathbb{P}(X_{n-1} = s_{n-1}, \dots, X_0 = s_0) > 0$.

This theorem is of great importance in the area of study of Markov chains, and we refer the interested reader to [Chung, 1960, §1.2] for a detailed proof. this theorem establishes a bridge between the theory of Markov processes in general and the properties of linear algebra and matrix theory. One such interesting property is the use of eigenvectors and eigenvalues of the transition matrix used to define a stationary distribution, which is what we are ultimately building towards this section.

We have seen that the transition matrix P governs the behaviour and the changes that happen from one state of the Markov chain to the next. A natural follow up question arises from trying to understand the behaviour of the chain an arbitrary $k > 1$ steps in the future instead of a single step, such as if we wanted to calculate $\mathbb{P}(X_{i+k} = s_{i+k} | X_i = s_i)$. We first note that this multi-step transition probability can be simplified, assuming that the Markov chain is homogenous², to the following:

$$\mathbb{P}(X_{i+k} = s_{i+k} | X_i = s_i) = \mathbb{P}(X_k = s_k | X_0 = s_0)$$

which we denote as $p_{i,k}^{(k)}$. Because we can define the individual probabilities of transitioning from one state to another after k steps, it stands to reason that there would be a transition matrix to describe this behaviour. The theorem that establishes the way these relationships work is the Chapman-Kolmogorov theorem and equation, which we present now. The formulation of this theorem presented in this work is taken from [Žitković, 2010, §8.3].

²A Markov chain is homogenous if the probability of transitioning from one state to another remains constant as we increase n .

Theorem 2 (Chapman-Kolmogorov Theorem). *Let $n, m \in \mathbb{N}_0$ and $s_i, s_j \in \mathcal{S}$. Then, $p_{i,j}^{(n+m)}$, which is the probability that starting from state i at step n we end up in state j after m steps is given by:*

$$p_{i,j}^{(n+m)} = \sum_{k \in \mathcal{S}} p_{i,k}^{(n)} p_{k,j}^{(m)}.$$

Proof. We will prove the case when the Markov process is homogenous. We begin by noting that $p_{i,j}^{(n+m)} = \mathbb{P}(X_{n+m} = s_j | X_n = s_i) = \mathbb{P}(X_m = s_j | X_0 = s_i) = p_{i,j}^{(m)}$. Applying the law of total probability and using the Markov property,

$$\begin{aligned} p_{i,j}^{(n+m)} &= \\ &= \mathbb{P}(X_{n+m} = s_j | X_n = s_i) \\ &= \mathbb{P}(X_m = s_j | X_0 = s_i) \\ &= \sum_{k \in \mathcal{S}} \frac{\mathbb{P}(X_m = s_j, X_n = s_k, X_0 = s_i)}{\mathbb{P}(X_0 = s_i)} \\ &= \sum_{k \in \mathcal{S}} \mathbb{P}(X_m = s_j | X_n = s_k) \mathbb{P}(X_n = s_k | X_0 = s_i) \\ &= \sum_{k \in \mathcal{S}} p_{i,k}^{(n)} p_{k,j}^{(m-n)}. \end{aligned}$$

□

We now introduce the following lemma that takes advantage of this property to provide an alternative way of computing and understanding the transition matrix P .

Lemma 1. *Let P^k be the k -th order matrix power of the transition matrix P for a homogeneous Markov chain $\{X_n, n \in \mathbb{N}_0\}$. Then,*

$$p_{i,j}^{(k)} = (P^k)_{i,j} \text{ for } i, j \in \mathcal{S}.$$

There is a way to prove this lemma by induction directly, but we opt instead to use the Chapman-Kolmogorov Equation $n-1$ times to achieve the final result, as follows.

This lemma becomes useful in proving the following theorem.

Theorem 3. *Let $n, m \in \mathbb{N}_0$ and $i, j \in \mathcal{S}$. Then, $p_{i,j}^{(n+m)}$, which is the probability that starting from state i in step n we end up in state j after m steps is given by:*

$$p_{i,j}^{(n+m)} = \sum_{k \in \mathcal{S}} p_{i,k}^{(m)} \cdot p_{k,j}^{(n)}.$$

From here, we now present two definitions relating to the properties of matrices which will become important when further exploring the types of Markov chains we are interested in.

Definition 4. *A matrix $A \in \mathbb{R}^{n \times m}$ is called non-negative if $a_{i,j} \geq 0$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$.*

Before continuing, we note that with stochastic matrices being just glorified assortments of probabilities that are neatly ordered, they are always non-negative.

Definition 5. *Let A be a non-negative matrix. If there exists $N_0 \geq 1$ such that all elements of A^{n_0} are positive, then we call A a quasi-positive matrix.*

These two definitions are necessary building blocks to get to the property we are truly interested in, ergodicity.

Definition 6. *A Markov chain $\{X_n : n \in \mathbb{N}_0\}$ is called ergodic if the limit $\pi_j := \lim_{n \rightarrow \infty} p_{i,j}^{(n)}$ exists for all $j \in \{1, 2, \dots, k\}$, is positive and does not depend on i , and $\pi = (\pi_1, \pi_2, \dots, \pi_k)$ is a probability distribution on $\mathcal{S}$.*

Ergodicity is a powerful property that does not arise intrinsically within the theory of stochastic processes, but is instead inherited from measure theory. Informally, a system is ergodic if the probability laws governing it remain invariant or unchanged by transitions between states or, in the case of time-indexed processes, by the passage of time. As a concrete illustration, consider tossing a fair coin 100 times. The distribution of outcomes observed by a single person performing all 100 tosses is, on average, identical to that obtained from 100 independent people each tossing the coin once; the underlying probability law is the same in both cases, and this system is ergodic. By contrast, suppose the coin is replaced by a biased one after 50 tosses. The probability law now changes mid-process and it is therefore not invariant across time or states which means that the system is therefore non-ergodic.

A thorough treatment of ergodicity lies beyond the scope of this work; the interested reader is directed to Sections 24 and 36 of [Billingsley, 2012] for a rigorous development through the lenses of measure theory and stochastic processes, respectively. At its core, ergodicity formalizes the equivalence between time averages and ensemble averages: rather than observing many independent realizations of a system simultaneously, one can instead observe a single realization over a long period of time and recover the same statistical quantities. This equivalence is precisely what makes ergodic Markov chains computationally tractable. The result that makes this precise, and upon

which the theoretical justification for Markov Chain Monte Carlo (MCMC) methods rests, is the Ergodic Theorem, stated below.

Theorem 4. *Let $\{X_n : n \in \mathbb{N}_0\}$ be an ergodic Markov chain such that*

$$\pi_j = \lim_{n \rightarrow \infty} p_{i,j}^{(n)}, \quad \text{for all } j \in \mathcal{S}.$$

Then the vector π is the unique solution of

$$\pi^T = \pi^T P,$$

and π is a probability distribution on $\mathcal{S}$.

Proof. Let j be an arbitrary element of $\mathcal{S}$. By the Chapman–Kolmogorov equations (Theorem 3),

$$p_{i,j}^{(n+1)} = \sum_{k \in \mathcal{S}} p_{i,k}^{(n)} p_{k,j}.$$

Taking the limit as $n \rightarrow \infty$ and applying $\pi_k = \lim_{n \rightarrow \infty} p_{i,k}^{(n)}$ for each $k \in \mathcal{S}$ gives

$$\pi_j = \sum_{k \in \mathcal{S}} \pi_k p_{k,j}.$$

Since $j \in \mathcal{S}$ was arbitrary, this holds for all j , which is precisely the component-wise form of $\pi^T = \pi^T P$.

To verify that π is a probability distribution: since $p_{i,j}^{(n)} \geq 0$ for all n , we have $\pi_j \geq 0$ for all $j \in \mathcal{S}$. Furthermore, since $\sum_{j \in \mathcal{S}} p_{i,j}^{(n)} = 1$ for all n , taking the limit yields $\sum_{j \in \mathcal{S}} \pi_j = 1$. Hence π is a valid probability distribution on $\mathcal{S}$.

Uniqueness follows from the ergodicity assumption: an ergodic (irreducible and aperiodic) chain admits exactly one stationary distribution, so π is the unique solution of $\pi^T = \pi^T P$. □

If we construct a Markov chain in a way that it is ergodic and control the distribution π so that it is something that we want, we can guarantee that after a sufficiently large amount of iterations n we will be sampling from the stationary distribution of the process, which is invariant through time because of the property presented in the above theorem. This is the crux of understanding the Markov Chain Monte Carlo technique which is the central object of study of this section. Further diving into the construction of the Metropolis–Hastings algorithm and the fundamental building blocks

needed to construct these chains that eventually lead to stationary distributions [we care about](#) would make this work exceedingly long, but [\[Rubinstein and Kroese, 2016\]](#) and [\[Robert and Casella, 1999\]](#) are good resources for a detailed rundown of these topics. We will return much later in this work when discussing the BASS method to MCMC and why it is so essential to this process. We will not look at pure MCMC though but an interesting modification to the base algorithm known as the Reversible Jump Markov Chain Monte Carlo (RJMCMC) algorithm.

For now though we move on to the other interesting sub-case of stochastic processes relevant to this work, the continuous-time Gaussian Process.

1.3 Gaussian Processes

A Gaussian process can be thought of as a generalization of the Gaussian (normal) distribution when viewed from the lens of stochastic processes. Stochastic processes, as we saw before, are indexed by a parameter, which itself belongs to an index set. The Gaussian process extends the definition of the Gaussian distribution from being valued on scalars or vectors to a more general index set of the form $T \subseteq \mathbb{R}^+$. Note that this set can contain an infinite number of points, which is the main difference between the Gaussian distribution and the Gaussian process. This extension is not trivial though, and to specify the structure of these particular types of stochastic processes, we aid ourselves with finite dimensional distributions. We now provide a formal definition of these types of distributions.

Definition 7. *Let $\{t_i\}_{i \leq k}$ be a sequence of points³ such that $\{t_i\}_{i \leq k} \subseteq T$. The k -dimensional random vector $(X_{t_1}, X_{t_2}, \dots, X_{t_k})$ has a distribution $\mu_{t_1, t_2, \dots, t_k}$ over $\mathbb{R}^k$. These measures $\mu_{t_1, t_2, \dots, t_k}$ are the finite dimensional distributions of the process.*

One logical follow up is to ask if the inverse of this definition is also true. This would be, starting from a consistent⁴ joint distribution of the random variables $(X_{t_1}, X_{t_2}, \dots, X_{t_k})$, with $k \in \mathbb{N}$, then would it be possible to define a probability measure $\mu_{t_1, t_2, \dots, t_k}$ on a higher cardinality set, in this case $\mathbb{R}^k$ that matches these distributions exactly? The positive answer to this question is the Kolmogorov Extension Theorem, an incredibly deep and powerful result

³Sometimes referred to as time points, considering this index not as an abstract set but a representation of the progress of time as indexed by a subset of the positive reals.

⁴Consistency is a fuzzy term with a lot of meanings, but its meaning in this context will become clear with the theorems that follow.

that guarantees the existence of stochastic processes by only defining a finite set of distributions that share some consistency requirements.

Theorem 5 (Kolmogorov Existence Theorem). *Let T denote some interval, and let $n \in \mathbb{N}$. For each $k \in \mathbb{N}$ and finite sequence of distinct times $t_1, \dots, t_k \in T$, let $\nu_{t_1 \dots t_k}$ be a probability measure on $(\mathbb{R}^n)^k$. Suppose that these measures satisfy two consistency conditions:*

1. *For all permutations π of $\{1, \dots, k\}$ and measurable sets $F_i \subseteq \mathbb{R}^n$,*

$$\nu_{t_{\pi(1)} \dots t_{\pi(k)}} (F_{\pi(1)} \times \dots \times F_{\pi(k)}) = \nu_{t_1 \dots t_k} (F_1 \times \dots \times F_k);$$

2. *For all measurable sets $F_i \subseteq \mathbb{R}^n$, $m \in \mathbb{N}$,*

$$\nu_{t_1 \dots t_k} (F_1 \times \dots \times F_k) = \nu_{t_1 \dots t_k, t_{k+1}, \dots, t_{k+m}} \left(F_1 \times \dots \times F_k \times \underbrace{\mathbb{R}^n \times \dots \times \mathbb{R}^n}_m \right).$$

Then there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a stochastic process $X : T \times \Omega \rightarrow \mathbb{R}^n$ such that

$$\nu_{t_1 \dots t_k} (F_1 \times \dots \times F_k) = \mathbb{P}(X_{t_1} \in F_1, \dots, X_{t_k} \in F_k)$$

for all $t_i \in T$, $k \in \mathbb{N}$, and measurable sets $F_i \subseteq \mathbb{R}^n$, i.e., X has $\nu_{t_1 \dots t_k}$ as its finite-dimensional distributions relative to times $t_1 \dots t_k$.

The Kolmogorov Extension Theorem addresses the question of whether we can construct a stochastic process given only a finite set of joint distributions. These joint distributions need to satisfy a consistency condition, which means that any finite subset of random variables should have joint distributions that are compatible with each other.

The theorem states that if such consistent joint distributions exist, then it is possible to define a stochastic process on a higher-dimensional space, often denoted as $\mathbb{R}^k$, such that the original given distributions match the marginal distributions of this process. In essence, the Kolmogorov Extension Theorem provides a way to create a stochastic process that captures the dependencies and correlations between random variables at different time points, even when we only have limited information about their joint distributions.

In the case of Gaussian processes, this means that if we have a finite set of random variables following consistent joint Gaussian distributions, we can construct a Gaussian process on a continuous domain (typically a subset of $\mathbb{R}$) such that any finite subset of the process is jointly Gaussian with the desired covariance structure. With this auxiliary result out of the way, we can now proceed to define the Gaussian process formally.

Definition 8. Let $\mathcal{X}$ be a nonempty index set, typically taken to be a subset of $\mathbb{R}^d$ for some $d \geq 1$. Let $\{X_t : t \in \mathcal{X}\}$ be a stochastic process. We call $\{X_t : t \in \mathcal{X}\}$ a Gaussian process if for every finite collection $\{t_i\}_{i \leq k} \subset \mathcal{X}$ and every finite k , the random vector $(X_{t_1}, X_{t_2}, \dots, X_{t_k})$ follows a multivariate Gaussian distribution when $k > 1$ and a univariate Gaussian distribution when $k = 1$.

An important extension of this is that we can interpret the Gaussian process as a random function f evaluated at input points $x \in \mathcal{X}$. That is, for each $x \in \mathcal{X}$, there is a random variable $f(x)$ representing the possible output of the function f at that location, or equivalently, a realization of a Gaussian distribution at x . This can also be understood as drawing a sample from the distribution of $f(x)$. As with any Gaussian distribution, $f(x)$ is completely characterized by its mean and variance. Unlike the finite-dimensional case, however, the index set $\mathcal{X}$ may contain infinitely many points, and so the natural approach is to characterize the process through a mean function and a covariance function defined over $\mathcal{X}$. We denote the mean function of the process as:

$$m(x) = \mathbb{E}[f(x)]$$

and the covariance function as:

$$k(x, x') = \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))].$$

The main reason why the covariance function is denoted by k is that this name alludes to the fact that k is a kernel of the terms in the input space. We now present a formal definition of this type of function, followed by some intuition on its interpretation. With these definitions though, we write the distribution function for f as

$$f(x) \sim \mathcal{GP}(m(x), k(x, x')).$$

Definition 9. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a nonempty input space, and let V be a vector space equipped with an inner product $\langle \cdot, \cdot \rangle_V$. Let $\varphi : \mathcal{X} \rightarrow V$ be a mapping, called a feature map. We call the function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ a kernel function if

$$k(x, x') = \langle \varphi(x), \varphi(x') \rangle_V, \quad \text{for all } x, x' \in \mathcal{X}.$$

What this kernel function allows us to do is essentially borrow an inner product from the feature space V and apply it to transformed elements of the input space $\mathcal{X}$. This allows us to compute inner products in a potentially high-dimensional or implicit feature space without requiring explicit knowledge of its structure. Tying the kernel back to Gaussian processes, we

note that while the mean function controls only the average value across realizations of f , the covariance function — that is, the kernel — governs the underlying behavior and structure of those realizations. Different choices of kernel encode different prior assumptions about the process, such as smoothness, periodicity, or stationarity. The rational quadratic kernel is a noteworthy case: it can be interpreted as a scale mixture of RBF kernels with different length scales, and interpolates between smooth and rough sample path behavior depending on the parameter α . Figure 1.2 illustrates how the choice of kernel affects the character of sample paths drawn from a Gaussian process.

Gaussian Process Samples Under Different Kernel Functions

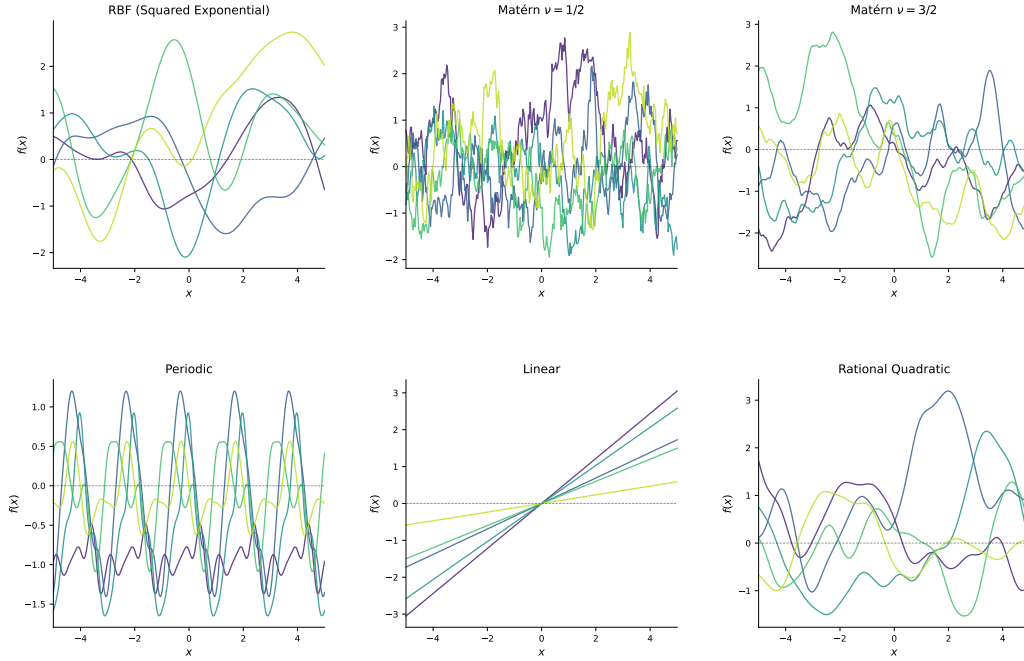


Figure 1.2: Sample paths from a zero-mean Gaussian process under six kernel functions: the RBF (squared exponential), Matérn $\nu = 1/2$, Matérn $\nu = 3/2$, periodic, linear, and rational quadratic kernels. Each colored trajectory represents an independent sample from the process.

Basically all major properties such as smoothness, differentiability, and range (to name only a few) are controlled by the choice of the kernel function. While the kernel function can be basically anything that fulfils the above [definition 9](#), there are some particularly useful or popular kernel functions when dealing with Gaussian processes. Chief among them is the squared

exponential kernel, also known as the radial basis function. This kernel is defined as

$$k(x, x') = \sigma_f^2 \exp \left(\frac{-\ell(x - x')^2}{2} \right),$$

where σ_f^2 and ℓ are hyperparameters. While we do not go into too much detail as to why this is a good choice for a kernel, we refer the interested reader to [Garnett, 2023].

The hyperparameters defined previously each control the general behaviour of the process when fitting data points. A more detailed run down of these points is given in the next section, which talks about Gaussian process regression. This is the method by which uncertainty is measured and function approximations are conducted in the traditional approach to Bayesian optimization.

1.4 Gaussian Process Regression

Gaussian process regression is the primary application of the theory developed in this chapter. It is a nonparametric approach to supervised learning in which, rather than fitting a fixed parametric model to data, one places a prior distribution directly over functions and updates it in light of observed data. Concretely, given a set of input-output pairs $\{(x_i, y_i)\}_{i=1}^n$, Gaussian process regression produces a posterior distribution over functions consistent with those observations, providing not only a point prediction at new inputs but a calibrated measure of uncertainty. It is this uncertainty quantification that makes Gaussian process regression particularly well-suited to Bayesian optimization, where the balance between exploration and exploitation depends critically on knowing where the model is uncertain.

The method generates from data a mean function and then uses that mean function to extrapolate or predict on points where there is no data observed. As we described in the last chapter, we can fully specify or determine a particular Gaussian process by specifying its mean $m(x) = \mathbb{E}[f(x)]$ and its kernel function $k(x, x')$. While in the last chapter we started with the Gaussian process and from that generate the values on the rest of the input space, in this chapter we are going to be doing somewhat of the inverse of that. We will be starting from a set of data $\{x_i\}_{i \leq n} = \mathcal{D} \subset \mathcal{X}$, and **computationally estimating** the **parameters** of the kernel and mean to achieve the best fit Gaussian process to the points and minimize the error.

One of the advantages of Gaussian process regression over other forms of supervised learning algorithms (such as linear regression) is the fact that Gaussian processes are a non-parametric model. Without delving too much

into the specifics, this means that the number of parameters controlling the model is not fixed at a certain quantity, but rather is adaptive. The amount of parameters can grow and shrink depending on the data that is being fit.

This however does not mean that we do not have control over the behavior of the function we decide to adjust to the points. In the following figure, we can see how changing the hyperparameter ℓ in the squared exponential covariance function changes the relative smoothness of the fit function.

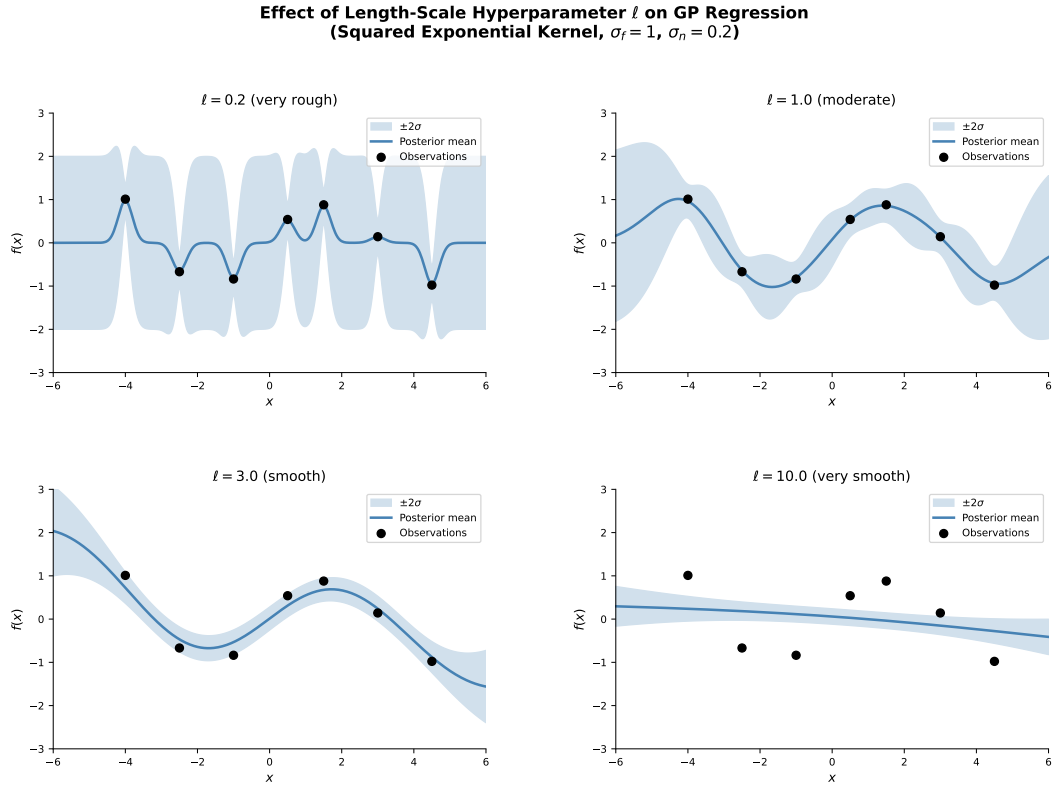


Figure 1.3: How different values of ℓ change the behaviour of the Gaussian process.

We also note how radically the confidence intervals change from one adjustment to the next based on moving this hyperparameter around. Because we can define an infinite amount of Gaussian processes based on our choice of hyperparameters (and further even by changing the kernel function), we require a principled criterion for selecting among them. The standard approach is to select the hyperparameters that best explain the observed data in a probabilistic sense, formalized as follows.

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be the observed data, and suppose observations are generated according to

$$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_n^2),$$

where $f \sim \mathcal{GP}(m, k_\theta)$ and θ denotes the vector of hyperparameters. Let $\mathbf{y} = (y_1, \dots, y_n)^T$ and let $K_\theta \in \mathbb{R}^{n \times n}$ denote the kernel matrix with entries $(K_\theta)_{ij} = k_\theta(x_i, x_j)$. Under the Gaussian process prior and the additive noise model, the marginal distribution of the observations is

$$\mathbf{y} \mid X, \theta \sim \mathcal{N}(\mathbf{0}, K_\theta + \sigma_n^2 I),$$

where we assume a zero mean function without loss of generality. Denoting $C_\theta = K_\theta + \sigma_n^2 I$, the log marginal likelihood is

$$\log p(\mathbf{y} \mid X, \theta) = -\frac{1}{2} \mathbf{y}^T C_\theta^{-1} \mathbf{y} - \frac{1}{2} \log \det C_\theta - \frac{n}{2} \log 2\pi.$$

The three terms admit a natural interpretation: the first is a data-fit term that penalizes poor prediction of the observations, the second is a complexity penalty that discourages unnecessarily flexible models, and the third is a normalizing constant. Maximizing the log marginal likelihood over θ is equivalent to the minimization problem

$$\hat{\theta} = \arg \min_{\theta} \left[\frac{1}{2} \mathbf{y}^T C_\theta^{-1} \mathbf{y} + \frac{1}{2} \log \det C_\theta \right], \quad (1.1)$$

where the constant term has been dropped as it does not depend on θ . This objective automatically balances data fit against model complexity, a property sometimes referred to as the *Occam's razor* effect of the marginal likelihood (more on this in §5.2 of [Rasmussen and Williams, 2006]). In the case of the squared exponential kernel, $\theta = (\ell, \sigma_f^2, \sigma_n^2)$, and (1.1) is typically minimized via gradient-based methods, as the gradient of the objective with respect to each hyperparameter can be computed analytically.

The minimization problem stated in (1.1) thus provides the formal criterion for hyperparameter selection: we seek the $\hat{\theta}$ that makes the observed data most probable under the model. This problem translates to finding the Gaussian process that best explains the observations in the sense that it minimizes residual uncertainty while penalizing unnecessary complexity. The solution involves selecting the optimal⁵ set of hyperparameters given

⁵Optimal is a dangerous word in this context. Depending on the problem and the structure of the marginal likelihood surface, the minimizer may not be unique, and in practice one often recovers a local rather than global minimum. Here we simply refer to the hyperparameters that solve (1.1), or a member of the family of solutions if multiple exist.

the observed data, and using the resulting mean function to predict values at unobserved inputs $\mathbf{x}_*$.

To arrive at this more concretely, we will begin by reviewing regression in general terms and then establish the formal link to Gaussian process regression, showing how the framework developed above gives rise to a fully specified supervised learning method.

1.4.1 Preamble to Gaussian Process Regression: A Re-hash on Regression Analysis

The central task of this section is regression, which is the process of estimating or generating understanding of a theoretical function $f : \mathcal{X} \rightarrow \mathbb{R}$ that describes the relationship between a response variable y and a group or single input variable described by the vector $\mathbf{x}$. While we do not know the exact behavior of f the function that describes this relationship⁶, what we do have is a set of observations $\mathcal{D} \subset \mathcal{X}$, where $\mathcal{D} = \{(x_i, y_i), i = 1, 2, \dots, n\}$ and $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. These observations constitute point-wise realizations of the function f . From that, for a set of new unknown points $\mathbf{x}_*$ we want to predict or infer the quantity $f(x_{i*}) = y_*$, which would be point-wise estimates of these unseen or unknown realizations of f .

One example of another method of solving this problem is the well known and well loved linear regression, that assumes that the function f is of the general form

$$y = f(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta} + \epsilon$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$. The error variance σ^2 and the coefficients $\boldsymbol{\beta}$ are generated by a process of minimizing the error of the outputs generated.

One of the disadvantages of [classical regression techniques such as linear regression](#) is that they only give one function as the output, but there may well be an infinite number of functions that fit a certain set of points. As an example, consider the following figure.

Notice how for the training dataset $\mathcal{D}$, all of these functions have error zero! The only recourse we have is to restrict these functions to generate one function (or at least a finite family of functions), and not an infinity of them. As with many other areas of mathematics, there is a trade off that we have to choose to perform between how general a solution to a problem is and how many solutions to that problem we wish to get.

The key insight to understanding Gaussian process regression from a statistician's point of view is changing the way we think about regression.

⁶We will discuss this in later chapters, but this is actually one of the greatest advantages of Bayesian Optimization, in that it can work on so called *black-box* functions.

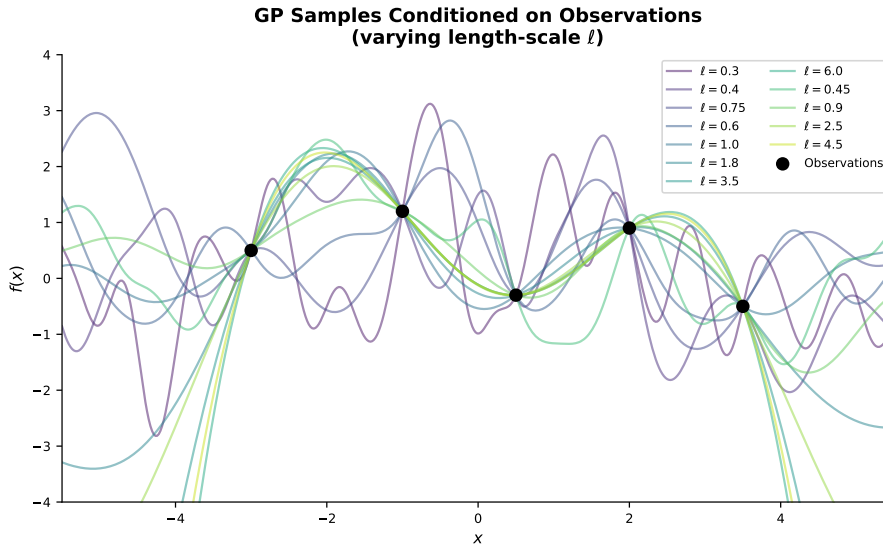


Figure 1.4: A sample of functions that could fit an arbitrary set of observations.

Instead of going the traditional way and providing restrictions to generate particular approximations, why don't we consider the space of possible functions that can fit these points as coming from a probability model? What if the observed functions in the past figure are realizations of a more general class of random variable that is coherent in the function space, rather than the discrete variable or vector space? From this groundwork is where Gaussian processes come into the picture. The only added restriction that we impose on the functions is that as a family, they come from a generalization to infinite dimensions of the multivariate normal distribution, and with that, we generate the entire conceptual framework from which we derive and generate the Gaussian process regression method. Having now set the expectation of what we wish to accomplish, we can now construct the theory of this different class of regression analysis.

1.4.2 Theory of Gaussian Process Regression (from the function space view)

This section of the text is heavily based on the work and structure of [Rasmussen and Williams, 2006] and to a much lesser extent [Wang, 2020]. The book Gaussian Processes for Machine Learning is the go-to reference and source for modern applications of Gaussian processes in general and it would be a disservice to the reader to not follow that structure and argumentation.

We can split up the set $\mathcal{D}$ into training and test, defining $\mathcal{D}_{train} = \{x_i, y_i\}_{i=1}^n$, and $\mathcal{D}_{test} = \{x_i^*, y_i^*\}_{i=1}^{n^*}$, with $x_i, x_i^* \in \mathbb{R}^d$ and $y_i, y_i^* \in \mathbb{R}$. As described before, what we want to achieve is basically generating out of sample predictions for the test set $\mathcal{D}_{test}$. Since the function $f = m$ defined is the mean of the process we defined before, we can simply evaluate the out of sample points x_i^* on f and we have an estimate for y_i^* . For simplicity and by convention, we denote the set of test outputs $\mathbf{f}_*$. Because of the structure we have defined, formally, the regression function f is really a normal distribution when conditioned by the training dataset $\mathcal{D}_{train}$, which would be

$$\mathbb{P}(\mathbf{f}|\mathbf{X}) = \mathcal{N}(\mathbf{f}|\boldsymbol{\mu}, k)$$

where $\mathbf{X} = [x_1, x_2, \dots, x_n]$, $\mathbf{f} = [f(x_1), f(x_2), \dots, f(x_n)]$, $\boldsymbol{\mu} = [m(x_1), m(x_2), \dots, m(x_n)]$ and k the kernel or covariance function of the process. By convention, it is customary to set $m(\mathbf{X}) = 0$, but there is a reasoning behind this. When dealing with the type of data generally modelled by these types of functions, it is common (encouraged even) to normalize the data before applying these methods to it. Another important piece of shorthand notation we will use is $K = k(\mathbf{X}, \mathbf{X})$, $K_* = k(\mathbf{X}, \mathbf{X}_*)$, and $K_{**} = k(\mathbf{X}_*, \mathbf{X}_*)$. Note that since the number of training points need not equal the number of test points, then n has not attributable relationship with n^* and we cannot state any properties of the matrix K_* , much less say that it is a square matrix.

Since we have formally defined the distribution of $\mathbf{f}$ and we know that $\mathbf{f}_*$ is the same distribution but considering a different set of data points $\mathbf{X}_* = [x_1^*, \dots, x_n^*]$, we can define the joint distribution of these vectors as

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} K & K_* \\ K_*^T & K_{**} \end{bmatrix} \right), \quad (1.2)$$

which is the joint probability distribution $\mathbf{f}, \mathbf{f}_* | \mathbf{X}, \mathbf{X}_*$. This is a massive step in constructing the regression model but we are still missing one component. When dealing with regression and prediction of values outside of the original training dataset, we do not need the joint probability, but the conditional probability of the test set given everything else we know. Formally, this means that we need to find the probability distribution of $\mathbb{P}(\mathbf{f}_* | \mathbf{X}, \mathbf{X}_*, \mathbf{f})$. Luckily, since we are dealing with Gaussian distributions, there is a very neat closed form expression that describes this probability. Referring to section A.2 of [Rasmussen and Williams, 2006], and noting that the mean for both $\mathbf{f}$ and $\mathbf{f}_*$ is 0, then

$$\mathbf{f}_* | \mathbf{f} \sim \mathcal{N}(K_*^T K^{-1} \mathbf{f}, K_{**} - K_*^T K^{-1} K_*).$$

This probability function is what is used to calculate out of sample predictions for the out of sample data $\mathbf{X}_*$. Concretely, the mean of this conditional distribution, $K_*^T K^{-1} \mathbf{f}$, is the point prediction we assign to $\mathbf{f}_*$ for each test point in $\mathbf{X}_*$. The intuition here is fairly straightforward, $K^{-1} \mathbf{f}$ can be thought of as a set of weights learned from the training data, and K_*^T projects those weights onto the test points by measuring how similar each test point is to each training point through the kernel k . Points in $\mathbf{X}_*$ that are close to training points in $\mathbf{X}$ will borrow more heavily from nearby observed function values, while points far from all training data will have their predictions pulled back toward the prior mean of zero.

Equally important is the second component, the covariance $K_{**} - K_*^T K^{-1} K_*$. This matrix tells us how uncertain we are about our predictions: K_{**} is the prior covariance between test points before seeing any data, and $K_*^T K^{-1} K_*$ is how much of that uncertainty we can explain away given the training observations. The diagonal entries of this matrix give the predictive variance at each test point, which we can use to construct credible intervals around our predictions. This is one of the more appealing properties of Gaussian process regression, that we get a full probability distribution over predictions rather than just point estimates.

Gaussian Process Regression Over Noisy Observations

When there is inherent randomness or noise in the signals being calculated, the structure of the problem itself changes. If we can assume that the noisy signals we are receiving are of the form $y = f(\mathbf{x}) + \varepsilon$ with the noise or inherent randomness captured by ε and these being independent, identically distributed observations of a normal or Gaussian distribution, the covariance [matrix](#) of the observations therefore becomes

$$\text{cov}(\mathbf{y}) = \sum \mathbf{y} = K(\mathbf{X}, \mathbf{X}) + \sigma^2 I,$$

with I the identity matrix. In practical terms this means that the inherent variance of every observation is present as an added term on the diagonal of the covariance matrix (this is because of the assumption that the noise is independent). This change causes the joint distribution originally defined in [equation 1.2](#) to

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K + \sigma^2 I & K_* \\ K_*^T & K_{**} \end{bmatrix} \right),$$

and the conditional distributions for prediction are therefore

$$\begin{aligned}\mathbf{f}_*|\mathbf{X}, \mathbf{y}, \mathbf{X}_* &\sim \mathcal{N}(\bar{\mathbf{f}}_*, \text{cov}(\mathbf{f}_*)), \text{ where} \\ \bar{\mathbf{f}}_* &:= \mathbb{E}[\mathbf{f}_*|\mathbf{X}, \mathbf{y}, \mathbf{X}_*] = K_*^T[K + \sigma^2 I]^{-1}\mathbf{y} \\ \text{cov}(\bar{\mathbf{f}}_*) &= K_{**} - K_*^T[K + \sigma^2 I]^{-1}K_*.\end{aligned}$$

These equations are for the entire training and test set though. For a test point x_* , we can define the vector $\mathbf{k}_*$ as the vector of covariances between this new test point and each of the points in the training dataset. This would be equivalent to taking the column associated with that single point from the test data matrix K_* . If we wanted just the closed form for this single out of sample observation, that would be

$$\begin{aligned}\bar{f}_* &= \mathbf{k}_*^T(K + \sigma^2 I)^{-1}\mathbf{y}, \\ \mathbb{V}(\bar{f}_*) &= k(x_*, x_*) - \mathbf{k}_*^T(K + \sigma^2 I)^{-1}\mathbf{k}_*.\end{aligned}$$

1.5 Summary and Outlook

We have covered a lot of ground in this chapter, moving from the discrete steps of Markov chains to the continuous, infinite-dimensional world of Gaussian processes. While these might seem like separate branches of probability, they provide the two pillars that support the rest of this work. The ergodic properties of the Markov chains we established are what will eventually allow us to actually run the BASS models through sampling based inference.

Gaussian process regression on the other hand gives us a way to turn a set of noisy observations into a full distribution of functions. As we have seen, the real power of the GP framework is that it doesn't just give us a single solution, rather it gives us a mathematically rigorous way to quantify what we don't know through the predictive variance.

However, having a good model is only half the battle. If we are dealing with a function that is expensive to evaluate, simply knowing its shape isn't enough—we need a plan. We need to know exactly where to look next to find the optimum as efficiently as possible. In the next chapter, we will take the posterior distributions we built here and use them as the engine for Bayesian optimization, moving from just modelling the function to making informed, sequential decisions about where we should evaluate it.

Chapter 2

Bayesian Optimization

This chapter (and the rest of this work) borrows heavily from the book [Garnett, 2023] as a source of consistent terminology. This book might be the most approachable and complete treatment of Bayesian optimization the author has found. We rely on it for standardizing notation and for some of the proofs and concepts, as well as for a general sketch of how this chapter is set out.

2.1 Introduction

Bayesian optimization is central to the experimental framework developed in this work. Understanding its motivation, assumptions, and mechanics is therefore essential before proceeding, and this section aims to provide a clear and self-contained foundation. The chapter that follows prioritizes conceptual clarity over generality and formal definitions with algorithmic details follow in subsequent chapters.

The necessity for Bayesian optimization arises from a particular kind of constraint, one that is distinct from the structural constraints we typically encounter in classical optimization. In traditional optimization, constraints govern the *form* of the solution space: for instance, in integer programming, the objective function is defined only over discrete inputs, or we restrict our interest to solutions that are integer-valued. In these settings, the primary objective is to locate an optimum of some objective function f over a well-defined domain $\mathcal{X}$. The computational cost of doing so, measured in the number of function evaluations required, is a secondary concern¹.

¹Sometimes this is not even a concern. We evaluate these functions in terms of the amount of wall-clock time it takes to converge to a solution, not necessarily the amount of computations required to arrive at that solution.

Bayesian optimization shares this goal but introduces a critical additional assumption: the objective function f is *expensive to evaluate*. That is, each query $f(\mathbf{x})$ carries a significant cost: computational, financial, physical, or temporal. In this scenario, exhaustive or gradient-based search over $\mathcal{X}$ becomes impractical. Under this paradigm minimizing the number of evaluations is no longer a secondary concern but a core objective in its own right. We therefore seek not only to find the optimum of f , but to do so in as few evaluations as possible.

To make this concrete, consider the problem of hyperparameter optimization in machine learning, which will become directly relevant to this work in later chapters. When training a neural network, one must specify a learning rate η which is the rate at which the model updates its weights in response to the gradient of the loss. If η is set too high, training may become unstable and fail to converge. If it is set too low, convergence will be prohibitively slow or the model may stagnate at a poor local minimum. Evaluating a single candidate value of η requires a full training run, which for modern large-scale models can take hours or days. Exhaustive grid search or random search over the hyperparameter space is therefore wasteful and sometimes impossible. A more principled approach is needed.

Bayesian optimization provides exactly this. Rather than treating f as a black box to be sampled blindly, we maintain a probabilistic *surrogate model* that encodes our current beliefs about the shape of f given the evaluations observed so far. At each iteration, an *acquisition function* uses this surrogate to identify the point in $\mathcal{X}$ that is most worth evaluating next, balancing the competing demands of *exploitation* (querying near known good regions) and *exploration* (querying in regions of high uncertainty). This loop composed of update the surrogate, optimize the acquisition function, evaluate f , and finally start again, continues until a convergence criterion is met or a budget of evaluations is exhausted.

There are many domains beyond hyperparameter tuning where this paradigm is natural and necessary: experimental design in the physical sciences, A/B testing in product development, materials discovery, and the calibration of expensive simulation models, among others. What these problems share is the same fundamental structure: an objective value or function that is cheap to specify but costly to evaluate, with no available gradient information and a need to reach a good solution within a limited query budget.

The remainder of this chapter develops the mathematical machinery required to make this framework precise. We begin with Gaussian processes as the surrogate model of choice, before turning to the construction and properties of standard acquisition functions, and finally to the full Bayesian optimization algorithm and its convergence behavior. As the models that

we build and the problems we solve in general become more and more complicated, there has been a growing interest in this area of statistics as a solution to generating viable extrema candidates of expensive to evaluate functions. There has been extensive work on this field recently and in the past few decades, with examples such as [Bergstra et al., 2013b], [Balandat et al., 2020], and [Akiba et al., 2019] that have impulsed this field considerably. Hyperparameter optimization has recently been talked about as the “killer app” for Bayesian optimization, which revitalized interest in Gaussian Processes in the context of machine learning and Bayesian optimization in general. The interest in the field has lead to a great amount of interest and therefore research and publications tackling the method and providing theoretical backing to it.

In this chapter we will show the basic theoretical structure of Bayesian optimization, the three central components that make it up, and the decisions we have to make to be able to approximate these functions from a Bayesian point of view.

2.2 Components of BO

Before describing the components of Bayesian optimization in detail, it is worth being precise about the class of functions we are concerned with. A function $f : \mathcal{X} \rightarrow \mathbb{R}$ is said to be a *black-box function* if its internal structure is either unknown or computationally infeasible to observe directly. More precisely, we have no access to an analytical expression for f , no gradient information $\nabla f(\mathbf{x})$, and no closed-form characterization of its global behavior. The only information available to us is the result of directly querying the function: given an input $\mathbf{x} \in \mathcal{X}$, we may observe the output $f(\mathbf{x})$. This stands in contrast to the functions typically encountered in classical optimization, where gradient-based methods can exploit the analytic structure of f to navigate the search space efficiently. In the black-box setting, no such structure is available, and evaluations themselves are the sole source of information about f .

Bayesian optimization is a sequential model-based approach to optimizing black-box functions that are expensive to evaluate. The goal is to find in some domain $\mathcal{X}$ a value x that maximizes the objective function f with the least possible number of evaluations. Practically, this means *evaluating* the value $f(x)$ for the least amount of times possible. One important distinction from other areas of optimization is that we do not impose any restrictions on the function f , except continuity². This is notably different

²As we will see in later chapters, the entire point of this work will be relaxing this

from many classical settings. In integer optimization for instance, the domain $\mathcal{X}$ is restricted to $\mathbb{Z}^d$, meaning the objective function is only defined (or only of interest) at discrete points, and the geometry of the search space changes fundamentally as a result. At the other extreme are methods such as the finite element method (FEM) used to solve partial differential equations numerically. They require not only that the functions involved be continuous, but that they belong to appropriate Sobolev spaces, possessing weak derivatives of sufficient order to guarantee the existence and stability of solutions. Bayesian optimization asks for neither: f need not be discrete, nor differentiable, nor analytically specified in any way except continuity. Formally, let f be the objective function that we want to maximize, and let $\mathcal{D}$ be the set of observations we have made so far, which consists of pairs (x_i, y_i) . We want to find the input x that maximizes the function $f(x)$, i.e., $x^* = \arg \max_{\mathcal{X}} f(x) = \arg \max_{\mathcal{X}} y$

The Bayesian optimization algorithm consists of three main components:

1. a prior distribution over the objective function,
2. an acquisition function,
3. and a method for updating the prior distribution based on the observed data.

The way this model works can be thought of as the following three steps:

1. Determining a probabilistic model that captures the observed behaviour of f , either by placing a prior distribution over f before any data has been observed, in the case of the first iteration, or by conditioning on accumulated data points $(x_i, f(x_i))$ for later iterations. We call this probabilistic model of f the prior, since it is used as one under this scheme³.
2. Optimizing the selected acquisition function which is a criterion, defined formally below, that quantifies the utility of evaluating f at a given point that probabilistically estimates the best possible points for evaluation in the following iteration.
3. Evaluating the function at the best candidate for improvement, therefore getting new data and updating our probabilistic model of the behaviour of the function.

restriction.

³Another common way to jump-start this process is to begin by evaluating some random points in $\mathcal{X}$ and continuing the process from there.

Each of these three steps is treated in detail in the sections that follow.

The prior distribution over the objective function $f(x)$ is usually assumed to be a Gaussian process, but that need not be the case. While these type of models are beyond the scope of this work, as examples, we refer to [Shah et al., 2013] for the prior set to a t -Student distribution and [Wang et al., 2022] for a case study of the Poisson prior. Given the set of observations $\mathcal{D}$, the selected prior distribution defines a posterior distribution $p(f \mid \mathcal{D})$, which is our updated belief over the objective function given the data observed so far that takes into account the uncertainty in the observations.

The acquisition function is a criterion that measures the expected utility of each point in the input space x for the purpose of optimization. The most commonly used acquisition function is the Expected Improvement (EI), which measures the expected improvement in the objective function if we were to evaluate the function at a particular input x . Much like the case with the prior distributions, there are other choices such as Thompson sampling (TS) and Upper Confidence bound (UCF), references for each can be found in [Kaufmann et al., 2012b] and [Kaufmann et al., 2012a] respectively.

$$EI(x) = \mathbb{E}[\max(f(x) - f(x_{best}), 0)],$$

where x_{best} is the current best input found so far. The idea is to select the input x that maximizes the EI, which balances exploration (trying new inputs) and exploitation (focusing on inputs that are likely to be good) this balance is controlled by hyperparameters, so that is another layer of complexity that needs to be controlled. In a sense, what we are doing is differing solving the main optimization problem by generating a simpler to solve optimization problem in finding the relevant extrema of the acquisition function, thus allowing us to spend as little function evaluations as possible on evaluating the target function f .

The final component of BO is the method for updating the prior distribution based on the observed data. This is done under the overarching philosophy of Bayesian inference, which allows us to compute the posterior distribution over the objective function given the prior distribution and the observed data. The posterior distribution is then used as the prior distribution for the next iteration of the algorithm. What we wish to do is infer the value $\phi = f(x)$ for some unobserved point in the domain $\mathcal{X}$.

2.3 Formalization

This process begins by setting the prior distribution $\mathbb{P}(\phi|x)$ which is our belief of possible values for ϕ before observation of the unknown x .

Assuming we observe the objective function at x and measure y , our optimization model supposes that the distribution of this measurement is determined by the value of interest ϕ through the observation model $\mathbb{P}(y|x, \phi)$. With this new value y we can update our beliefs and generate the posterior function through the use of Bayes' theorem:

$$\mathbb{P}(\phi|x, y) = \frac{\mathbb{P}(\phi|x) \cdot \mathbb{P}(y|x, \phi)}{\mathbb{P}(y|x)}.$$

As with all applications of Bayes' theorem such as this one, the denominator is simply a constant term used to scale the probability. This means that we can simplify the expression by only keeping the kernel of the distribution and dropping the normalizing term, and using the proportionality operator $\propto$, as follows.

$$\mathbb{P}(\phi|x, y) \propto \mathbb{P}(\phi|x) \cdot \mathbb{P}(y|x, \phi).$$

In the Bayesian framework, the posterior distribution is often not the final goal but a starting point for further tasks such as prediction or decision-making. This builds on the iterative process under which we are working in Bayesian optimization, where new information is added to the approximation of the function every time a function evaluation happens. As an example, consider the case where we want to predict the outcome of an independent, repeated noisy observation at x, y' . We can treat the outcome as a random variable and derive its distribution by integrating our posterior belief about ϕ against the observation model:

$$\mathbb{P}(y'|x, \mathcal{D}) = \int \mathbb{P}(\phi|x, \mathcal{D}) \cdot \mathbb{P}(y'|x, \phi) d\phi.$$

This equation represents the posterior predictive distribution for y' given we know x and a set of data $\mathcal{D}$. By integrating over all possible values of ϕ , weighted by their plausibility, the posterior predictive distribution naturally accounts for uncertainty in the unknown objective function value.

From here is where the link to Gaussian Processes lies. Gaussian Processes serve as a natural bridge between the Bayesian framework and Bayesian optimization. They provide a flexible, non-parametric prior of the form $\mathbb{P}(f)$ over a certain function f , and their posterior predictive distributions can be efficiently computed and used to guide the search for the optimal input value. The acquisition function, which is derived from the GP's posterior predictive distribution, plays a crucial role in balancing exploration and exploitation during the optimization process.

Acabo aqui o faltaria algo más?

Chapter 3

Sequential Model Based Optimization

There has been a lot of work done over the last few years in the space of Bayesian Optimization. One primary advance has been the formalization of sequential model based optimization (SMBO) as a general framework for Bayesian Optimization. The general idea of these types of models is to divide the optimization problem into two stages, first fitting a model from a specific class that predicts the performance on the target function and then gathering additional data to further refine the model that is used as a surrogate for the actual function we wish to evaluate. As detailed in [Koehrsen, 2018], the two steps can be broken down into this pseudo-algorithm for the problem of finding the best hyperparameters for a generic machine learning model:

1. Build a surrogate probability model for the target function
2. Find the hyperparameters that perform the best on the surrogate
3. Apply these hyperparameters to the true objective function¹
4. Update the surrogate model incorporating the new results
5. Repeat steps 2-4 until a limit is reached (maximum number of iterations, time spent, or others such as cost).

While all SMBO algorithms follow these basic steps, the true power of it is in the flexibility it provides when selecting the surrogate model and when selecting how the new results can be incorporated into the model itself (steps 3 and 4). **expand on this, it is the main point of the thesis.**

¹This is the part that is expensive to evaluate, ideally, we want to minimize the amount of times that we actually run the true objective function.

As is clear by the title of this work, the main contribution is the implementation of the BASS model as a surrogate model under this framework, but there have been a large amount of different methods and models used as surrogate models as well as a large amount of ways of incorporating the results into the surrogate, known as the selection function or evaluation criteria depending on the literature being read. The next sections detail the other models that are commonly used as surrogate models, and where these models got their roots.

3.1 A Survey of Existing Surrogate Models

Different surrogate models can capture different aspects of the underlying function, each with its own strengths and weaknesses. The most common choice is Gaussian Processes (GPs). The selection depends on factors such as the dimensionality of the problem, the smoothness of the objective function, its domain (fully continuous, hybrid or discrete), and the available computational resources. In this section, we provide some texture as to the common surrogate models for the framework.

3.1.1 Gaussian Processes

While much time has already been dedicated to GPs in this work, it is worth noting that the strengths that define this method merit it. First and foremost, the models provide a robust probabilistic framework under which to model the underlying function. They not only predict the mean of the function but also provide a measure of uncertainty baked in (variance and covariance) associated with the predictions outside of the already estimated points. This uncertainty estimation is what allows Bayesian Optimization to work, since the selection of the next best point to evaluate is generally the point that we have the most uncertainty about, coupled with it being a proper candidate for a minima or maxima, depending on the situation. The choice between selecting candidates based on uncertainty or chances of being a better choice than the already explored points give us a trade-off in that we can tune depending on what we find most relevant for the problem by adjusting the models hyperparameters.

While these models do not work well for combinatorial domains² because these models explicitly assume a continuous search space, if the problem

²This is a very loaded statement, but the authors think it is correct. There is also literature to back this up, such as [Wang et al., 2023] §3.2.

has a relatively low dimension (not uncommon for hyperparameter optimization), then it is very efficient computationally to calculate. There is also the matter of kernel selection, which gives the user unparalleled flexibility in continuous domains, since the kernel can be adapted to predict in a family of functions (such as strictly linear ones) if there is information known about the behaviour of the objective function already known.

Another advantage that cannot be overstated is the interpretability of these Gaussian Processes. These models not only provide mean value calculations, but also confidence intervals. This makes it easier to explain why the model is making certain choices, and with most types of problems, it is generally preferable to start with a simple and understandable method and scale up if necessary than to use a highly complex method which might become its own black box. Because of the nature of these models, they also tend to work well with small amounts of data, which goes hand in hand nicely with the restriction that we set off with for Bayesian Optimization, that being that the objective function is expensive to evaluate. This becomes a problem when dealing with surrogate models such as neural networks, which require orders of magnitude more data to make proper predictions.

Yet another large benefit worth discussing is the fact that these models interpolate. The black box objective functions that we are assigning uncertainties to and calculating pointwise values for are at the completely deterministic. What the framework of Bayesian Optimization allows us to do is assign that uncertainty based on our own standing relative to the function, this is, the functions themselves are not probabilistic or uncertain, but our understanding of them is. Other surrogate models such as BASS which we are proposing do not interpolate perfectly, in the case of BASS because the estimation is done through MCMC which generates a probability distribution over a point in this use case. Because of their underlying construction and properties Gaussian Processes not only interpolate with zero variance in the points already calculated, but assign a low variance to regions of the domain close to the ones that have been calculated, and as such reduce the need for unnecessary exploration which is important under the framework that the objective functions are costly to evaluate.

To conclude, it is important to note that while all of these theoretical benefits are incredibly important, it is also worth mentioning that there is a lot of work in existing libraries and frameworks that has already been done to facilitate the use of GPs in the context of Bayesian Optimization. Because it is the de-facto standard in the area, most all of the existing packages for SMBO and BO specifically include modules and tools to define a Gaussian Process as the surrogate model. Practical inertia must not be discounted as a critical reason to explain the prevalence of GPs in this field.

3.1.2 Tree Parzen Estimators

The inception of this algorithm comes from the paper [Bergstra et al., 2011a], which provides a thorough explanation of the mechanisms that define TPEs (Tree Parzen Estimators). Even though this algorithm was created in 2013, it is still considered one of the premier algorithms for hyperparameter tuning using BO. This algorithm excels at problems with high dimensionality and small fitness evaluation budgets. The success of TPEs lies in their ability to efficiently explore the hyperparameter space by modelling the objective function using a combination of tree-based estimators and probabilistic density estimation techniques, such as Parzen estimators (also referred to as kernel density estimators), which entail a straightforward averaging of kernels centered around existing data points. Probably the most interesting aspect of this surrogate model is that while we normally are interested in calculating $\mathbb{P}(y|x)$ which is the probability of an objective function given the points already calculated, TPE takes a different approach and calculates $\mathbb{P}(x|y)$, which is the probability of having the points already calculated given a certain objective function being the “correct” one.

By iteratively updating a probabilistic model of the objective function and using it to guide the search for promising hyperparameter configurations, TPEs strike a balance between exploration and exploitation, allowing it to converge to near-optimal solutions with relatively few evaluations. One large benefit to TPE models is that there is only one hyperparameter to tune, namely $\gamma \in [0, 1]$, which controls what we consider to be a good function evaluation and what we consider to be a bad one. Going into more detail than this would require a section onto itself, but more details can be found in [Bergstra et al., 2011a] and in the paper for Hyperopt, which is the python library published by the creators of the model for hyperparameter tuning in python [Bergstra et al., 2013b]. One important thing to consider is that these types of models require more data to achieve great results when compared to GPs in general, and as such, if the function is prohibitively expensive, it could become a problem.

3.1.3 Bayesian Neural Networks

Bayesian neural networks (BNNs) are a type of neural network that incorporates Bayesian methods to model uncertainty in the network’s weights and predictions. Traditional neural networks typically output a point estimate for each input, representing a deterministic prediction. In contrast, BNNs provide a probabilistic framework that quantifies uncertainty associated with predictions. Bayesian neural networks, with their probabilistic framework,

provide a natural way to model the objective function and its uncertainty. During the optimization process, BNNs can be trained on the observed data points (input-output pairs), and the resulting posterior distribution over the weights can be used to make predictions about the objective function’s behaviour in unobserved regions of the input space.

Furthermore, BNNs offer the advantage of not only providing point estimates of the objective function but also quantifying the uncertainty associated with those estimates. This uncertainty estimation is crucial in BO for guiding the exploration-exploitation trade-off.

This area of research is thriving, and many different neural network architectures have been proposed to further optimize the BO model. This work does not delve into any particular methodology, but a run-down of recent advances and their particular use cases (e.g., large scale data, high dimensional BO, combinatorial optimization) can be seen in [Wang et al., 2023].

3.1.4 Random Forest, Gradient Boosted Machines

Tree and forest based algorithms have received a lot of praise in recent years. They are unreasonably effective when dealing with tabular data in a traditional setting, even outperforming deep neural networks which seem to have overtaken these models in everything but tabular data [Grinsztajn et al., 2022]. In the context of BO, they can be good surrogate models, as they are generally good regressors. They are highly flexible, as they can automatically handle nonlinearities, interactions, and high-dimensional feature spaces without the need for explicit feature engineering. They are also highly scalable, as they can handle a wide range of problem sizes and complexities, making them suitable for BO tasks with varying levels of computational resources.

While these models may seem like a natural choice, there are some limitations in selecting them as surrogate models, most relevantly, the lack of probabilistic outputs. Unlike Bayesian Neural Networks (BNNs) or GPs, RFs and GBMs do not naturally provide probabilistic outputs or uncertainty estimates. This makes it more challenging to quantify uncertainty in predictions, which is crucial for guiding the exploration-exploitation trade-off in BO. This is lessened by the fact that they can indirectly capture uncertainty through techniques such as bootstrapping and out-of-bag estimation, and as such, these methods can be leveraged to estimate the variability in predictions and guide the exploration-exploitation trade-off in BO. There are also some other limitations, like the fact that these models are basically black boxes in and of themselves if the problem is of a high dimension or there are many interactions between variables, or that if we are not careful with regularization, they can be overfit very easily. One other large problem is simply the amount

of hyperparameters these models have, as tuning them to perform well on a BO problem is not an easy task, and changes between one objective function and another can make previous hyperparameter configurations unusable, or lead to model stagnation, which is not something we want with objective functions that are expensive to evaluate.

A practical advantage that cannot be overlooked is the sheer amount of libraries and support that exists for these types of models. They have been studied intensely for the last decades, and the library implementations that exist in popular programming languages are some of the best designed Machine Learning libraries out there. Still, the choice of a RF or GBM must be done carefully, as heavy trade-offs could hurt more than help in the long run.

3.2 Other Techniques for Optimizing Hyperparameters

3.2.1 Grid Search

In grid search, the hyperparameter space is divided into a grid, and models are evaluated at each point in the grid. While grid search is exhaustive and guaranteed to find the optimal solution within the search space. The largest trade-off is that this method of search is expensive, since each function evaluation is defined by the problem to be expensive. If this is not a problem, then this is one of the best methods out there. The constraint of expensive to evaluate black-box functions is what necessitates the rise of other search algorithms, such as Bayesian Optimization.

3.2.2 Random Search

This is a simple technique where hyperparameters are randomly selected from a predefined search space. Despite its simplicity, random search can often be surprisingly effective and computationally efficient. One of the main problems comes when the search space becomes complex, because of the curse of dimensionality, using a random search strategy could lead to a very narrow search in terms of the space as a whole. This can be compounded by not looking at points “close” to the ones where we already found good results, which can lead to us not finding local or global maximums or minimums, and instead spending valuable time looking at regions we might already know to probably not yield interesting results. A good reference for both grid search methods and random search can be found in [\[Bergstra and Bengio, 2012\]](#).

3.2.3 Genetic Algorithms

Part of the larger group composed of evolutionary algorithms, they are based on the idea of natural selection when evaluating certain points in the hyperparameter mesh that we define. To extend the analogy, the fittest (in this case the hyperparameter vectors with the lowest or highest values) are selected to produce offspring. This is done iteratively, and these individuals' traits (good guesses for the optimums) are passed on to the next generations. These algorithms are generally more complex than anything else covered in this work and as such we do not elaborate much further, but a good and recent reference is [Xiao et al., 2020], which describes these algorithms used to tune hyperparameters in deep learning models.

Chapter 4

BASS and their Origins

The main proposal in this work is the use of a BASS model as the surrogate model in the framework of Bayesian Optimization rather than a traditional Gaussian Process or any of the alternatives already discussed. To understand this method and why it is useful for this purpose, we develop first the theory that lead to the construction of this method, and then explain its behaviour in some detail.

As with most other algorithms used for regression tasks, BASS is built upon a rich history of work that was previously done in the field. It begins with CART and decision trees, which MARS models are based on, and in turn BMARS (Bayesian MARS) uses the same model form as MARS but builds the model using the traditional Bayesian approach of defining a prior and constructing from there. Finally, BASS is a specific case of the BMARS model. This is much more clear when viewed as a diagram, as shown in the following figure.



Figure 4.1: The theoretical basis for BASS.

4.1 Decision Trees

Decision trees are a fundamental and versatile tool that comes in two main flavours: classification trees and regression trees. These two types of trees serve distinct purposes and are employed to make predictions in different contexts. The term "Classification and Regression Tree" (CART) analysis serves as an umbrella term that encompasses both classification and regression tree procedures. CART was first introduced by Breiman et al. in 1984 [Breiman, 2017], and it represents a unified framework for decision tree modelling, allowing for flexibility in handling both classification and regression tasks within the same algorithm. While classification and regression trees share common principles, such as recursive binary splitting and impurity measures (e.g., Gini impurity for classification and mean squared error for regression), they also exhibit some differences in terms of the criteria used to determine the best feature and split point at each node. The final predictions for a new data point are obtained by traversing the tree from the root node to a leaf node and returning the average (for regression) of the target variable for the training samples in that leaf node.



Figure 4.2: A decision tree.

In classification tree analysis, the primary objective is to predict the discrete class or category to which a given data point belongs. The decision tree algorithm partitions the data into subsets based on input features and assigns each data point to a specific class by following a path through the tree.

On the other hand, regression tree analysis is employed when the predicted outcome is a continuous, real-valued number. Similar to classification trees, regression trees partition the data based on input features. However, in this context, the goal is to estimate a numeric value for each data point, typically by computing the average of the target variable within each leaf node.

In both cases, the decision tree serves as a transparent and interpretable model, making it valuable for exploratory data analysis and predictive modelling. The tree's structure allows for a straightforward understanding of how decisions are made, and predictions for new data points are obtained by navigating the tree from the root node to an appropriate leaf node, where the final prediction is based on the characteristics of the training samples within that leaf.

The versatility and comprehensibility of decision trees have contributed significantly to their widespread use in various fields, and the distinctions between classification and regression trees make them adaptable tools for addressing a wide range of data analysis and prediction tasks. One problem that these methods tend to have though is overfitting, since if we allow the model to keep generating new nodes or leafs, the tree ultimately becomes too complex and every data point that we would like to analyse is simply too narrow to be useful as an aggregation scheme.

While decision trees are useful on their own their real power comes from the versatility they provide, leading to a wide range of methods being born from the basic idea of splitting decisions into paths. These methods are sometimes called *ensemble* methods, because they involve the creation and combination of multiple models to form a more robust and accurate predictive model. As an interesting side note, the term ensemble is borrowed from the world of music and theatre. Instead of relying on a single predictive model, these models create an ensemble of multiple models, each of which may have its own strengths and weaknesses. These individual models, often referred to as base models or weak learners, are combined in a way that leverages their collective intelligence to make better predictions.

Ensemble methods work on the premise that combining diverse models can reduce the risk of overfitting, enhance generalization, and improve predictive accuracy. Each model in the ensemble may make errors on some portions of the data, but by aggregating their predictions, the ensemble can produce more accurate and stable results. Common ensemble methods include Random Forest, Gradient Boosting, Adaptive Boosting, Bagging, and many others. The difference in these models is how they combine these different decision trees to achieve a more stable model.

4.2 MARS Models

Now, let's shift our focus the following technique down the chain for us known as MARS, which stands for Multivariate Adaptive Regression Splines. MARS builds upon the foundation laid by decision trees but introduces a differ-

ent paradigm for modelling complex relationships within data. Rather than using piecewise constant approximations, MARS leverages piecewise linear functions, thereby extending the toolbox of non-linear regression methods.

MARS is particularly valuable when the relationships between input features and the target variable are not strictly linear but can be approximated as a series of linear segments. Let's delve into the mathematical foundations of MARS to gain a deeper understanding.

For the following section, we will adopt the terminology and follow the development presented in [Friedman, 1991a] and [Friedman, 1991b].

MARS is defined using functions that are piecewise linear in the form of $(x - t)_+$ and $(t - x)_+$. These functions take the maximum between 0 and the value within the function, as follows:

$$(x - t)_+ = \max\{0, x - t\} = \begin{cases} x - t & \text{if } x > t, \\ 0 & \text{if } x \leq t. \end{cases}$$

Each function is piecewise linear with a slope change at the value t , which makes them linear splines. Each pair divided at the value t is referred to as a “reflected pair”. The idea of the method is to form reflected pairs for each input variable X_j with slope changes a selection of observed values x_{ij} . Following this construction and considering a variable X_j , the collection of basis functions is:

$$\mathcal{C} = \{(X_j - t)_+, (t - X_j)_+\}, \text{ with } t \in \{x_{1,j}, x_{2,j}, \dots, x_{N,j}\}, \text{ } j = 1, 2, \dots, p.$$

And each reflected pair appears as follows, and are shown in figure 4.3:

$$h_1(X) = h(X_j - x_{ij})$$

$$h_2(X) = h(x_{i,j} - X_j)$$

So, for each X_j , the set of candidate reflected pairs that exist at each of the x_{ij} points of that variable is shown in figure 4.4.

This means that if all input values are distinct, there are a total of $2Np$ basis functions, and $2N$ divisions in each of the splines for each variable. The model we use to combine all variables is an additive one in β :

$$f(X) = \beta_0 + \sum_{m=1}^M \beta_m h_m(X),$$

where each function $h_m(X)$ is an element of $\mathcal{C}$ or a linear combination of these basis functions. For this initial explanation, we will use only functions that

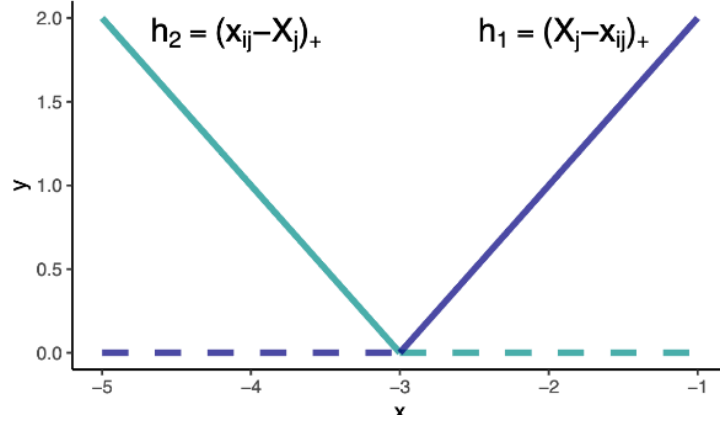


Figure 4.3: The constructed reflected pairs.

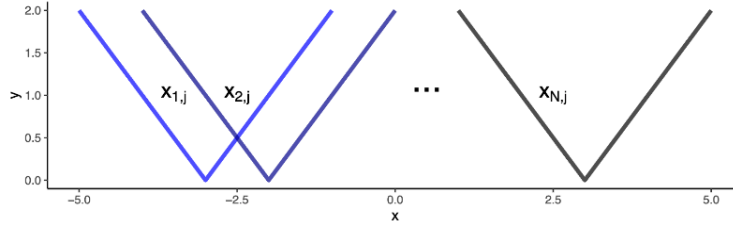


Figure 4.4: Collections of hinge functions.

do not involve interactions (the model with interactions works very similarly to CART decision trees).

The model construction process (denoted as $\mathcal{M}$) begins with the base model $\hat{f}(X) = \hat{\beta}_0 = h_0(X) = 1$, where a term functioning as an intercept at the origin is defined. To add terms incrementally, all functions in the set $\mathcal{C}$ are candidates for entry into $\mathcal{M}$. For each observation x_{ij} , the new model involving these functions is fitted as follows:

$$\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 h_1(X) + \hat{\beta}_2 h_2(X).$$

Since this is a linear form in terms of each of the functions h_i , the adjustment of each parameter $\hat{\beta}_i$ is done by minimizing the sum of squares.

The reflected pair that is added is the one that minimizes the training error. This process is repeated for each of the remaining reflected pairs, solving OLS to determine the new β , and adding the one that continues to minimize the error. The stopping criterion can be either that none of the remaining reflected pairs reduces the error enough (in absolute or relative terms) or until we have a predetermined number of variables in the model.

This leads to overfitting the data, but this is the goal in this part of the procedure when minimizing the training error. Once we have $\mathcal{M}$, we start the second part of model fitting, which is the pruning part. In this case, terms $h_i(X)$ are removed iteratively, starting with the one that produces the smallest increase in the residual sum of squares when removed. This procedure yields a better model for each λ size, denoted as $\hat{f}_\lambda$.

Several procedures can be used to estimate the optimal value of λ , such as cross-validation or bootstrap, but this involves a significant computational cost. To minimize this cost, MARS models generally use a generalized cross-validation (GCV) procedure. This criterion is defined as:

$$GCV(\lambda) = \frac{\sum_{i=1}^N (y_i - \hat{f}_\lambda(x_i))^2}{\left(\frac{1-M(\lambda)}{N}\right)^2},$$

where the value $M(\lambda)$ is the number of effective parameters in the model, depending on the number of terms plus the number of breakpoints used, penalized by a factor (this factor is 2 in the case for additive in X that we are explaining, and 3 when there are interactions).

Now that we have described the base model, we can go back and consider the differences when incorporating interaction terms. Instead of just adding terms additively, products between the existing h_ℓ functions in the model and between each reflected pair of each x_{ij} can be added. Note that this general case encompasses the previous one because if you want to add a reflected pair without interacting with the other variables, you simply multiply by $h_0 = 1$.

In this way, a model $\mathcal{M}$ with M terms will be updated as follows after finding the combination $\hat{\beta}_{M+1}h_\ell(X) \cdot (X_j - t)_+ + \hat{\beta}_{M+2}h_\ell(X) \cdot (t - X_j)_+$ that best minimizes the training error:

$$\hat{f}(X) = \hat{\beta}_0 + \sum_{m=1}^M \hat{\beta}_m h_m(X) + \hat{\beta}_{M+1} h_\ell(X) \cdot (X_j - t)_+ + \hat{\beta}_{M+2} h_\ell(X) \cdot (t - X_j)_+,$$

where $h_\ell \in \mathcal{M}$ and $t = x_{ij}$ for any case. **Como transiciono de una idea a otra aqui?**

One interesting consideration is that the added flexibility of these models does not end with it being piecewise linear, as one small alteration that can be made is to replace the truncated linear functions with truncated cubic basis functions or other functions of a higher order, which provides allows the resulting model to be fully differentiable within its domain.

In conclusion, our exploration of Multivariate Adaptive Regression Splines (MARS) is a powerful and versatile technique for modelling complex relationships within data. Unlike traditional linear regression, MARS leverages

piecewise linear functions, making it a valuable tool when dealing with non-linear relationships between input features and the target variable. The methodology behind MARS involves the creation of reflected pairs and the incremental addition of basis functions to the model, effectively adapting to the data's intricacies. The model construction process aims to minimize the training error by iteratively selecting the reflected pair that provides the most significant reduction in error. This leads to a model that can potentially overfit the data, but this is by design during the training phase. Subsequently, a pruning step is employed to refine the model, removing terms that do not contribute significantly to the model's predictive power.

4.3 BMARS Models

The key innovation that distinguishes BMARS from its predecessor lies in its Bayesian framework. While MARS primarily focuses on constructing piecewise linear models by iteratively selecting basis functions to minimize training error, BMARS introduces Bayesian methods to bring a probabilistic perspective into this process. This integration of Bayesian principles allows BMARS to provide not only point estimates of model parameters but also entire probability distributions, enabling a more comprehensive understanding of the uncertainty inherent in the modelling process.

As described in [Denison et al., 1998b], which is the original paper where this method was proposed, the idea behind this model is to allow more flexibility in the traditional MARS model by looking at it from a Bayesian perspective. This is done by mimicking the MARS model with changes in framing, namely by considering the number of basis functions, along with their *type*, their coefficients and their form (the positions of the split points and the sign indicators) as random.

We take the definition of a basis type from the original paper and a small example from there to make the definition tangible, and include it here for completeness.

Definition 10. *We consider basis functions X_i and X_j to be of the same type if $[x(1, i), \dots, x(J_i, i)]$ is identical to some permutation of $[x(1, j), \dots, x(J_j, j)]$. Hence, with m predictor variables, there are N different types of basis functions, where N is given by:*

$$N = \sum_{i=1}^m \binom{m}{i} = 2^m - 1. \quad (4.1)$$

Note that the sum does not include the constant basis function term (for which i would equal 0 in (4.1)), as this basis X_1 is always the sole constant

basis function in the model, so it cannot be chosen as a candidate basis. Frequently, some maximum order of interaction I is assigned to the model, such that $J_i \leq I$ for $i = 1, \dots, k$. In which case, the sum in (4.1) only runs from 1 up to I . We let $T_i \in \{1, 2, \dots, N\}$ denote the type of basis function X_i ; thus, T_i effectively tells us which predictor variables we are splitting on, i.e., what the values of $x(1, i), \dots, x(J_i, i)$ are.

As an example, suppose we have a problem with just two predictors, i.e., $m = 2$. Then the types of basis functions that could be in the model (not including the constant one) are $[\pm(x_1 - *)]$ (say, type 1), $[\pm(x_2 - *)]$ (say, type 2), and $[\pm(x_1 - *)][\pm(x_2 - *)]$ (say, type 3). So for all those basis functions which split only on predictor x_2 , their types T_i are all equal to 2.

The key insight that this framework of construction is based on is that any MARS model can be uniquely defined by the following:

- The number of basis functions k ,
- the coefficients of the basis functions a_i ,
- the type of basis functions T_i ,
- the knot points associated with each interacting term t_{ji} ,
- the sign indicators associated with each interacting term s_{ji} .

and as such, a probability distribution over the possible MARS models can be constructed, enabling a more comprehensive understanding of the uncertainty inherent in the modelling process since the response is an entire probability distribution rather than a single model. This framework also provides the ability to do ANOVA and Sobol decompositions for sensitivity analysis. One important aspect of this framework for construction is that the number of parameters varies with the number of basis functions k , and as such, a traditional method such as Markov chain Monte Carlo cannot be used directly, as it assumes constant parameters. An extended version of MCMC is therefore used, called reversible jump MCMC (first introduced in [Green, 1995b]), that allows for a variable number of parameters. *Cual es el modelo? Tengo el paper pero no lo veo claro.*

4.4 BASS Models

Bayesian Adaptive Spline Surface (BASS) models are an extension of the family of non parametric regression models of which multivariate adaptive regression splines (MARS) was the first developed in 1991 and is the most

basic [Friedman, 1991b]. MARS is an extension of linear models, where interactions between variables and non linearities are modelled automatically.

The extension of this model of interest is the BMARS model. As the authors state in [Denison et al., 1998b], the aim of BMARS is to provide a Bayesian model that mimics the MARS procedure. This is done by considering the number of basis functions, along with their type, the coefficients and their form, random. Then, a suitable MCMC algorithm, in this case reversible jump monte carlo [Green, 1995b], is used to calculate a suitable posterior for the problem. The great advantage to RJMCMC over other MCMC techniques is that the number of parameters can be variable, and as such, is a great basis for the type of analysis that is necessary for models in the MARS class.

BASS models are an adaptation of BMARS models that include some efficiency improvements for posterior sampling alongside parallel tempering for better posterior exploration [Francom and Sansó, 2020b]. For our purposes though, the most significant improvement is that it allows the use of categorical variables alongside some continuous variables. This is incredibly useful in the context of hyper-parameter optimization, where there are generally some categorical variables such as polynomial degree, smoothing, etc. While we remain in the one dimensional case and this improvement is not necessarily useful, the multi dimensional case will benefit greatly from this improvement.

A detailed rundown of the model, its priors, and its construction in general can be seen in [Francom and Sansó, 2020b], but, we include the definition of the model presented in that work for completeness.

Let y_i denote the dependent variable and $\mathbf{x}_i$ denote a vector of p independent variables, with $i = 1, \dots, n$. Without loss of generality, let each independent variable be scaled to be between zero and one. We model y_i as

$$y_i = f(\mathbf{x}_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

$$f(\mathbf{x}) = a_0 + \sum_{m=1}^M a_m B_m(\mathbf{x})$$

$$B_m(\mathbf{x}) = \prod_{k=1}^{K_m} g_{km} [s_{km}(\mathbf{x}_{v_{km}} - t_{km})]_+^\alpha$$

The only difference between this setup and that of MARS and BMARS is the inclusion of the constant g_{km} in each element of the tensor product. This term is $g_{km} = [s_{km}(\mathbf{x}_{v_{km}} - t_{km})]_+^\alpha$, where $s_{km} \in \{-1, 1\}$ is referred to as the sign, $t_{km} \in [0, 1]$ is a knot, v_{km} selects a variable, K_m is the degree

of interaction, and α determines the degree of the polynomial splines. This normalizes the basis functions so that the basis coefficients $a_1, \dots, a_M$ are on the same scale, making computations more stable.

4.5 BASS Models as Surrogates in Bayesian Optimization

With the discussions of the previous chapter and this one, we now have the tools to use the BASS model as a surrogate. There are a couple of decisions that we have to make though so that the model works properly, discussed in detail in this section. Namely, they are the starting points of the problem, the discretization of the function domain, and how we will handle uncertainty in points that have not yet been evaluated.

4.5.1 Starting Point for the Algorithm

One common problem with BO is the cold start problem. Basically, it is very difficult for the system to draw inference or make predictions at the beginning of iterations because it simply has no relevant information. Another way of saying this is that the prior in this case is uniform (or discrete uniform depending on the context) over the entire function domain, which is not good. A simple solution to this problem is to give the algorithm a set of starting points in the domain, this could be to either do a small grid search to start for a small number of iterations, or do a random search with the same objective. Having these few points, then the surrogate has enough information to properly decide a good next point to evaluate with the specific acquisition function selected. The next chapter concerns the experiments that were run to test the potency and viability of this model, and in it, different amounts of starting points are used. Needing the starting points is not something unique to BASS models though, as many other models (for example GPs) also have trouble with cold starts. For the experiments, we made sure to begin all of them on the same amount of already evaluated points, calculated at random.

4.5.2 Discretization of the Function Domain

As with any numerical problem, one important step in designing the algorithm to solve it is the discretization of continuous variables. This process involves dividing the continuous input space into discrete intervals or points, enabling the algorithm to operate efficiently within a finite computational

framework. The choice of discretization scheme can significantly impact the performance and convergence properties of the optimization process.

One approach to discretization is to define a grid over the input space. This involves partitioning each dimension into a set of equally spaced points, forming a mesh that covers the entire domain. While grid-based discretization offers simplicity and uniformity, it will suffer from the curse of dimensionality, especially in high-dimensional spaces, where the number of grid points grows exponentially with dimensionality.

The way we decided to discretize the space was using Latin Hypercube Sampling (LHS) [McKay et al., 2000]. LHS is a statistical sampling method used to generate a set of diverse and representative samples from a multidimensional probability distribution. It is particularly useful in optimization, where obtaining a set of samples that cover the entire parameter space evenly and efficiently is crucial. This method is better for this purpose than a simple grid search or a traditional pseudo-random sample of points generated through Monte Carlo methods. We do not go into too much detail into these methods as they are not the main point of this work, but the interested reader can look to the previously referenced [McKay et al., 2000] or [Stein, 1987].

4.5.3 Uncertainty Estimation

One problem in selecting the BASS method as the surrogate compared to GPs is that it does not have a built in mechanism for determining the uncertainty of points outside of the ones already evaluated. This means that there is no real mechanism to use alongside the acquisition function to select the next best point to evaluate, the literal point of Bayesian Optimization. This might seem like a method-breaking problem, but thankfully, there are options to calculate this uncertainty. The approach we decide to take is a relatively simple one, where the uncertainty in the points that we have is directly proportional to how close it is to other points that have already been estimated. This means that at the end of the day, the sample space for the possible next points in the sequence (if they are continuous) are the midpoints between each value already evaluated. The weight that we give to these points is tunable as to give the user some control over exploration in the model or experimentation in regions already decided to be good, as will be seen in the next chapter with the hyperparameter choices of κ and σ .

Chapter 5

Experimental Validation

5.1 Introduction

The previous chapters established the theoretical basis for Bayesian Optimization (BO) with BASS models as surrogates. We discussed initialization challenges, domain discretization, and uncertainty estimation, with special attention to unexplored regions of the search space. The next step is empirical validation.

This chapter presents the experimental framework used to evaluate BASS-based BO for hyperparameter optimization. The goal is to compare BASS against established alternatives and determine where it performs well, where it underperforms, and why. In particular, we study whether BASS can address practical concerns such as cold starts, sample efficiency, and reliable optimization in high-dimensional settings.

The methods compared in this chapter are Grid Search, Random Search, Gaussian Process-based BO (GP-BO), and BASS-based BO (BASS-BO). Experiments are conducted across multiple models, datasets, and synthetic benchmark functions to provide a broad evaluation. The analysis focuses on both optimization quality and computational efficiency.

The chapter is organized as follows. First, we describe the experimental design, including methods, models, datasets, metrics, and statistical procedures. Next, we present and interpret the results. Finally, we summarize the main findings, identify limitations, and provide recommendations for future work.

5.2 Experimental Design

5.2.1 Objective and Scope

The primary objective is to evaluate BASS-BO against established hyperparameter optimization methods on two dimensions: optimization effectiveness and optimization efficiency.

This evaluation criterion is consistent with the central motivation of BO and Sequential Model-Based Optimization, namely to reduce expensive objective function calls while maintaining strong optimization performance. The study includes classical benchmark functions (functions from [Surjanovic and Bingham, 2024]) to test behavior under controlled and application-oriented conditions.

5.2.2 Hyperparameter Optimization Methods

The methods compared are the ones previously mentioned in this work along with the new framework with BASS as the surrogate:

- **BASS-Based Bayesian Optimization (BASS-BO):** the proposed method. BASS surrogates model the response surface and uncertainty, and an acquisition function selects new evaluations iteratively.
- **Random Search:** random sampling from predefined hyperparameter distributions. It is often more efficient than Grid Search in moderate to high dimensions, as shown in [Bergstra and Bengio, 2012].
- **Gaussian Process-Based Bayesian Optimization (GP-BO):** a standard BO baseline using Gaussian Processes for surrogate modeling and uncertainty-guided acquisition.

The dataset in particular we will be using is the Virtual Library of Simulation Experiments: Test Functions and Datasets [Surjanovic and Bingham, 2024]. This suite is composed of standard functions such as the Branin-Hoo function, commonly used to test Bayesian Optimization problems.

5.2.3 Experimental Procedure

Each run follows a common protocol to ensure fair comparison.

1) Initialization All methods start from the same initial design generated by random sampling. This reduces cold-start bias and ensures that early information is matched across methods.

2) Iterative optimization

- **BASS-BO**: fit BASS surrogate on observed trials, optimize acquisition function, evaluate selected candidate, update surrogate, repeat.
- **GP-BO**: same loop with GP surrogate.
- **Random Search**: sample and evaluate candidates independently each iteration.
- **Grid Search**: evaluate according to grid ordering until budget is exhausted or grid is completed.

3) Budget and stopping A fixed evaluation budget is used per task to make efficiency comparisons meaningful. When a method terminates early (for example, finite grid exhausted), remaining iterations are treated as no additional improvement.

4) Repetitions and randomization Each configuration is repeated across multiple random seeds. Method order and data split assignments are randomized to reduce systematic effects.

5.2.4 Evaluation

As for evaluation of the methods, the central trade-off at the heart of Bayesian Optimization is that we want to achieve great returns at the lowest possible cost. On those axes, we define a performance metric and some efficiency metrics as well to test out our methods. The performance metric is simply the best objective value reached within budget. As for the efficiency metric, we track the evaluations over time to see how the methods behave under different constraints.

5.2.5 Statistical Analysis

To compare methods rigorously, we use confidence intervals to make sure we have a wide understanding of the behavior of the methods themselves, as well as effect sizes to quantify practical significance.

5.2.6 Initial Testing Conditions

The initial testing conditions used in this study are listed below. Full implementation details, including scripts, targets, and output files, are available in the project repository: github.com/AdrianTJ/BayesianOptim_BASS.

- Branin-Hoo function ($2d$), bounded domain)
- Rasteguín ($4d$, bounded domain)
- Additional benchmark functions (to be added after completion of all runs)

All methods are compared under a fixed evaluation budget and identical initialization strategy. For each replicate, an initial design of size $n_0 = \max(2d + 1, 8)$ is generated via maximin Latin Hypercube Sampling (LHS) where d is the dimension of the particular problem, and each optimizer proceeds for a fixed number of iterations. Results are aggregated over multiple random seeds (in our testing, 25 of them), and convergence is summarized using the best-so-far objective value at each iteration. Particular care was taken for these function evaluations to ensure reproducibility, so the set seed always starts at the default value of 1001, and proceeds to seed 1026.

5.2.7 BASS-BO Hyperparameters and Practical Tuning Space

A central advantage of the proposed BASS-BO framework is that it exposes several tunable controls that directly affect exploration, exploitation, uncertainty calibration, and local refinement. In contrast to a more rigid surrogate setup, these controls allow the optimizer to adapt its behavior to the objective landscape and the available evaluation budget. In the current implementation, the most relevant BASS-BO parameters are:

- `bass_kappa_start`, `bass_kappa_end`: define the exploration schedule in the lower confidence bound (LCB) acquisition rule. A larger early value encourages broader search, while a smaller late value emphasizes refinement.
- `n_cand`: number of candidate points sampled per iteration for acquisition maximization/minimization. Larger values improve candidate quality but increase computation.
- `bass_local_frac`: fraction of candidates sampled locally around the current best point. Higher values increase local exploitation.
- `bass_local_sd`: local sampling spread around the current best point in normalized coordinates. Larger values widen local search neighborhoods.

- `bass_explore_every`: frequency of forced exploration steps. At this interval, candidate selection prioritizes posterior uncertainty rather than LCB score.
- `bass_degree_early`, `bass_degree_late`, `bass_switch_after`: control surrogate complexity over time. Lower early degree can improve stability with limited data, while higher late degree increases expressiveness once more observations are available.
- `bass_sd_inflate`: multiplicative inflation factor applied to posterior standard deviation for acquisition. This counteracts early overconfidence and increases exploratory behavior.
- `bass_sd_floor`: minimum uncertainty floor to prevent near-zero posterior variance from collapsing exploration.

These parameters make BASS-BO highly configurable. The same flexibility is also a source of sensitivity, since poor settings can degrade convergence and it can become easy to shoot oneself in the foot with so much freedom. For this reason, we treat parameter selection as part of the method design, and report both the defaults and tuned settings used in each experiment. All reported numbers are with the default parameter set in the code, presented below.

- `bass_kappa_start` = 3.5
- `bass_kappa_end` = 1.5
- `bass_sd_floor` = 10^{-3}
- `bass_sd_inflate` = 1.20
- `bass_local_frac` = 0.35
- `bass_local_sd` = 0.08
- `bass_explore_every` = 7
- `bass_degree_early` = 1
- `bass_degree_late` = 2
- `bass_switch_after` = 40

This configuration is intended to provide strong early exploration, gradual transition to exploitation, and robust behavior against early uncertainty collapse.

5.2.8 Interpretation of BASS-BO Controls

The BASS-BO controls can be grouped into four functional categories:

1. **Exploration–Exploitation Controls:** `bass_kappa_start`, `bass_kappa_end`, and `bass_explore_every`. These govern how aggressively the method explores uncertain regions versus exploiting known promising areas.
2. **Candidate-Set Design Controls:** `n_cand`, `bass_local_frac`, and `bass_local_sd`. These determine whether search is globally diverse, locally concentrated, or hybrid.
3. **Surrogate Complexity Controls:** `bass_degree_early`, `bass_degree_late`, and `bass_switch_after`. These control how quickly model complexity increases as data accumulates.
4. **Uncertainty Calibration Controls:** `bass_sd_inflate` and `bass_sd_floor`. These reduce overconfident behavior and improve robustness in early iterations.

Empirically, these groups interact. For example, increasing `n_cand` without revisiting `bass_kappa_*` may not yield better search behavior, and adding strong local sampling (`bass_local_frac`) can require stronger uncertainty inflation (`bass_sd_inflate`) to avoid premature convergence. An important type of test that should be run with these types of parameters is sensitivity analysis, but that is outside the reach of this current project.

5.3 Results

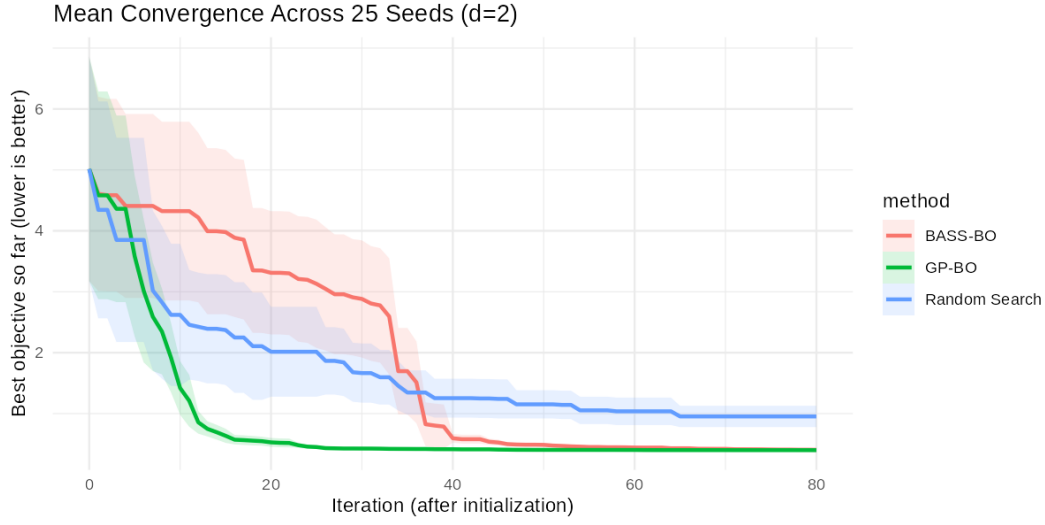
What follows are the graphs representing some runs from the BASS-BO cycle, compared to GP-BO and Random Search. We ran these optimizers across 10 different functions, as to get wider coverage into the actual performance of the optimizers.

5.3.1 Branin-Hoo Function

Formally, defined as:

$$f(x_1, x_2) = \left(x_2 - \frac{5.1}{4\pi^2} x_1^2 + \frac{5}{\pi} x_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos(x_1) + 10, \quad (5.1)$$

$$\text{with } x_1 \in [-5, 10], \quad x_2 \in [0, 15].$$

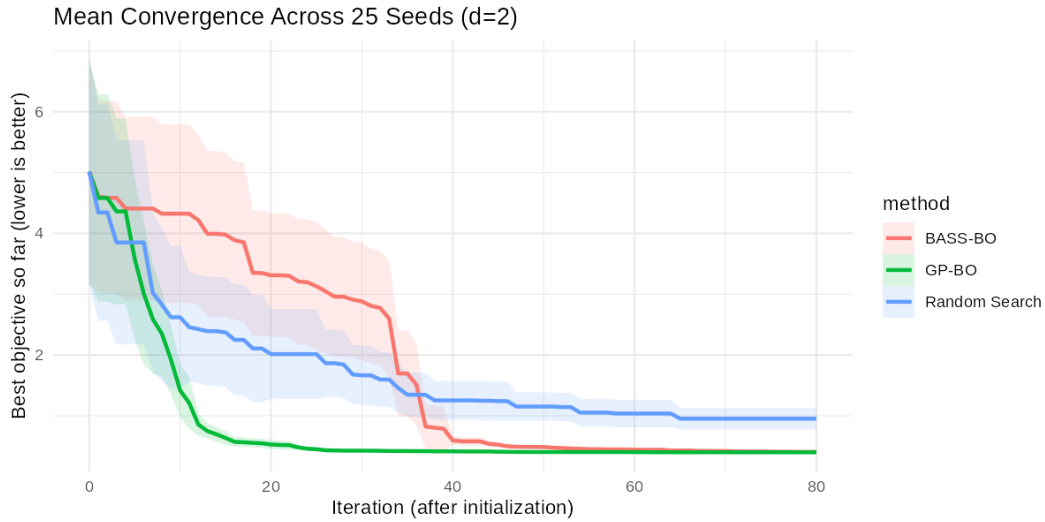


5.3.2 Rastrigin Function

$$f(\mathbf{x}) = An + \sum_{i=1}^n [x_i^2 - A \cos(2\pi x_i)], \quad (5.2)$$

$$\mathbf{x} = (x_1, \dots, x_n), \quad x_i \in [-5.12, 5.12], \quad i = 1, \dots, n,$$

where $A = 10$ is the standard choice. We used the 4 dimensional version of this function.



5.4 Result Interpretation and Limitations

Chapter 6

Conclusions and Future Work

6.1 Conclusions

This thesis studied the use of Bayesian Adaptive Spline Surfaces (BASS) as surrogate models within a Bayesian Optimization framework for black-box optimization and hyperparameter tuning. The central objective was to determine whether a BASS-based optimizer can offer competitive sample efficiency when compared with established baselines, while also providing practical flexibility for difficult objective landscapes. To evaluate this, a complete experimental pipeline was developed in R, including multi-dimensional optimization support, reproducible seed-based replication, parallel execution, and standardized convergence reporting against GP-BO and Random Search.

The empirical results show that all methods improve objective values as evaluations accumulate, which confirms that the implementation and evaluation protocol are functioning as intended. In the current completed benchmark setting, GP-BO converges faster than BASS-BO and reaches lower objective values earlier in the budget. Random Search improves more slowly, as expected, and is generally less sample-efficient than both surrogate-guided methods. BASS-BO remains competitive in final performance in several runs, but its average convergence speed is lower than GP-BO on the smooth low-dimensional cases completed so far. This pattern is consistent with the fact that GP surrogates are particularly well matched to smooth, low-dimensional benchmark functions such as Branin-Hoo.

A key conclusion of this work is that BASS-BO should not be interpreted as a universally superior replacement for GP-BO. Instead, it should be seen as a flexible alternative whose performance depends strongly on the optimization regime and on calibration of its control parameters. In this thesis implementation, BASS-BO exposes a richer set of

practical tuning knobs than the GP baseline, including dynamic exploration schedules, uncertainty inflation and flooring, local versus global candidate mixing, periodic forced exploration, and surrogate complexity scheduling. These controls are meaningful because they allow the optimizer to adapt its behavior over the course of the search. At the same time, they increase sensitivity to configuration choices, **so BASS-BO requires more careful tuning to achieve its best performance.**

From a methodological perspective, one contribution of this thesis is the framing of Bayesian Optimization as a configurable optimization system rather than a single fixed algorithm. The experiments demonstrate that decisions about candidate generation, uncertainty calibration, and exploration scheduling have measurable effects on convergence, sometimes as large as the differences between surrogate families themselves. This finding is important for practice, because it implies that fair comparisons between BO methods must hold the full experimental protocol constant, including initialization strategy, candidate pool size, duplicate handling, seed replication, and stopping budget. Without this consistency, measured performance differences can reflect procedure rather than model quality.

The current study also has scope limits. The strongest comparative evidence presently comes from a relatively small benchmark set. Because benchmark diversity is limited at this stage, the results should be interpreted as directional rather than universal. In particular, the present experiments emphasize smooth functions where GP assumptions are favorable. **The relative advantage of BASS-BO may increase on objectives that are more irregular, nonstationary, or interaction-heavy,** and this remains an open empirical question that the established codebase is now prepared to test.

The practical takeaway is therefore balanced. For smooth, low-dimensional objectives under tight evaluation budgets, GP-BO is an excellent default and currently outperforms BASS-BO in convergence speed in this thesis. However, BASS-BO remains a viable method, especially when one expects model mismatch under GP assumptions or when adaptive control of exploration behavior is desirable. Additionally, the BASS method naturally accepts categorical variables, so in those cases the model naturally scales to the required problem space. The additional BASS-BO parameters are not simply for show; they are mechanisms that can change search behavior in ways that may become advantageous on more difficult landscapes.

6.2 Future Work

Future work should extend this analysis in two directions. First, the benchmark suite should be expanded to include more multimodal and higher-dimensional objectives, including noisy settings, so that conclusions are less tied to any one function class. Second, a structured sensitivity study should be completed for the BASS-BO controls, with seed-matched statistical comparisons and effect sizes for both final objective value and convergence-area metrics. Together, these extensions would provide a stronger basis for identifying when BASS-BO is preferable to GP-BO and for recommending robust default configurations.

In summary, this thesis delivers a reproducible and extensible framework for comparing surrogate-based Bayesian Optimization methods, and demonstrates that BASS can be integrated effectively in a multi-dimensional BO loop. The current evidence favors GP-BO on the completed smooth benchmark cases, but it also shows that BASS-BO offers a richer control surface that may be valuable in harder optimization regimes. The main outcome is therefore both empirical and methodological: GP-BO is the stronger baseline in the tested setting, while BASS-BO emerges as a promising, highly tunable alternative whose full potential depends on problem structure and careful calibration.

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